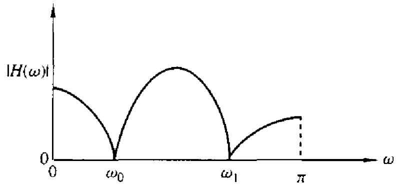


Figure 4.50 Frequency response characteristic of a notch filter.

To create a null in the frequency response of a filter at a frequency ω_0, we simply introduce a pair of complex-conjugate zeros on the unit circle at an angle ω_0. That is,

$$z_{1,2} = e^{\pm j\omega_0}$$

Thus the system function for an FIR notch filter is simply

$$\begin{aligned} H(z) &= b_0(1 - e^{j\omega_0}z^{-1})(1 - e^{-j\omega_0}z^{-1}) \\ &= b_0(1 - 2\cos\omega_0 z^{-1} + z^{-2}) \end{aligned} \tag{4.5.30}$$

As an illustration. Fig. **4.51** shows the magnitude response for a notch filter having a null at $\omega = \pi/4$.

The problem with the FIR notch filter is that the notch has a relatively large bandwidth, which means that other frequency components around the desired null are severely attenuated. To reduce the bandwidth of the null, we can resort to a more sophisticated, longer FIR filter designed according to criteria described in Chapter 8. Alternatively. we could. in an ad hoc manner, attempt to improve on the frequency response characteristics by introducing poles in the system function.

Suppose that we place a pair of complex-conjugate poles at

$$p_{1,2} = re^{\pm j\omega_0}$$

The effect of the poles is to introduce a resonance in the vicinity of the null and thus to reduce the bandwidth of the notch. The system function for the resulting filter is

$$H(z) = b_0 \frac{1 - 2\cos\omega_0 z^{-1} + z^{-2}}{1 - 2r\cos\omega_0 z^{-1} + r^2 z^{-2}} \tag{4.5.31}$$

The magnitude response $|H(\omega)|$ of the filter in **(4.5.31)** is plotted in Fig. **4.52** for $\omega_0 = \pi/4$, $r = 0.85$, and for $\omega_0 = \pi/4$, $r = \mathbf{0.95}$. When compared with **the** frequency response of the FIR filter in Fig. **4.51,** we note that the effect of **the** poles is to reduce the bandwidth of the notch.

In addition to reducing the bandwidth of the notch, the introduction of a pole in the vicinity of the null may result in a small ripple in the passband of the filter due to the resonance created by the pole. The effect of the ripple can be reduced by introducing additional poles and/or zeros in **the** system function of the notch filter. The major problem with this approach is that it is basically an ad hoc, trial-and-error method.

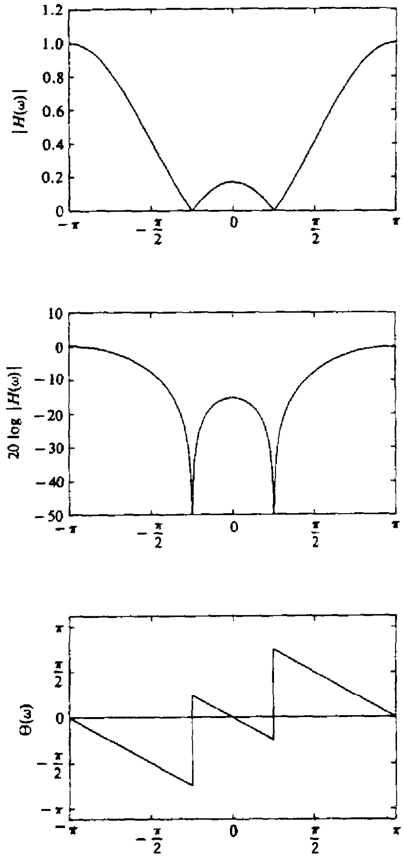


Figure 4.51 Frequency response charactenstics of a notch filter with a notch at $\omega = \pi/4$ or $f = 1/8$; $H(z) = G[1 - 2\cos\omega_0 z^{-1} + z^{-2}]$.

4.5.5 Comb Filters

In its simplest form, a comb filter can be viewed as a notch filter in which the nulls occur periodically across the frequency band, hence the analogy to an ordinary comb that has periodically spaced teeth. Comb filters find applications in a wide range of practical systems such as in the rejection of power-line harmonics, in the separation of solar and lunar components from ionospheric measurements of electron concentration, and in the suppression of clutter from fixed objects in moving-target-indicator (MTI) radars.

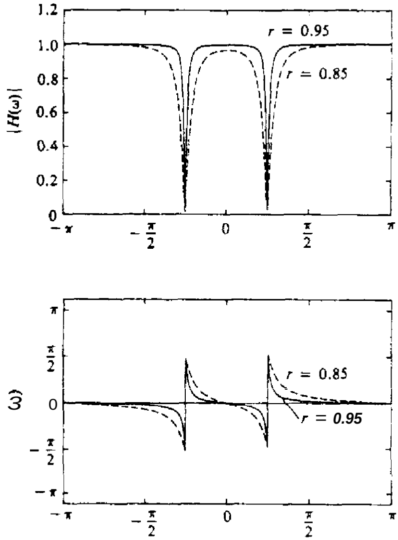


Figure 4.52 Frequency response characteristics of two notch filters with poles at (1) $r = 0.85$ and (2) $r = 0.95$; $H(z) = b_0[(1 - 2\cos\omega_0 z^{-1} + z^{-2})/(1 - 2r\cos\omega_0 z^{-1} + r^2 z^{-2})]$.

To illustrate a simple form of a comb filter. consider a moving average (FIR) filter described by the difference equation

$$y(n) = \frac{1}{M+1}\sum_{k=0}^{M} x(n-k) \tag{4.5.32}$$

The system function of this FIR filter is

$$\begin{aligned} H(z) &= \frac{1}{M+1}\sum_{k=0}^{M} z^{-k} \\ &= \frac{1}{M+1}\frac{[1-z^{-(M+1)}]}{(1-z^{-1})} \end{aligned} \tag{4.5.33}$$

and its frequency response is

$$H(\omega) = \frac{e^{-j\omega M/2}}{M+1}\frac{\sin\omega(\frac{M+1}{2})}{\sin(\omega/2)} \tag{4.5.34}$$

From (4.5.33) we observe that the filter has zeros on the unit circle at

$$z = e^{j2\pi k/(M+1)} \qquad k = 1, 2, 3, \ldots, M \tag{4.5.35}$$

Note that the pole at $z = 1$ is actually canceled by the zero at $z = 1$. so that in effect the FIR filter does not contain poles outside $z = 0$.

A plot of the magnitude characteristic of (4.5.34) clearly illustrates the existence of the periodically spaced zeros in frequency at $\omega_k = 2\pi k/(M+1)$ for $k = 1, 2, \ldots, M$. Figure 4.53 shows $|H(\omega)|$ for $M = 10$.

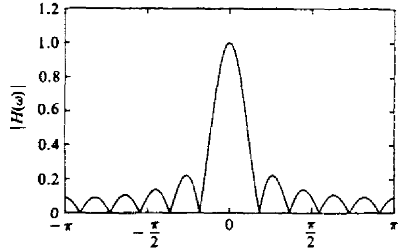


Figure 4.53 Magnitude response charactenstic for the comb filter given by (5.4.32) with $M = 10$.

In more general terms, we can create a comb filter by taking an FIR filter with system function

$$H(z) = \sum_{k=0}^{M} h(k)z^{-k} \tag{4.5.36}$$

and replacing z by z^L, where L is a positive integer. Thus the **new** FIR filter has **a** system function

$$H_L(z) = \sum_{k=0}^{M} h(k)z^{-kL} \tag{4.5.37}$$

If the frequency response of the original FIR filter is $H(\omega)$, the frequency response of the FIR in (4.5.37) is

$$\begin{aligned} H_L(\omega) &= \sum_{k=0}^{M} h(k)e^{-jkL\omega} \\ &= H(L\omega) \end{aligned} \tag{4.5.38}$$

Consequently, the frequency response characteristic $H_L(\omega)$ is simply an L-order repetition of $H(\omega)$ in the range $0 \le \omega \le 2\pi$. Figure 4.54 illustrates the relationship between $H_L(\omega)$ and $H(o)$ for L = 5.

Now, suppose that the original FIR filter with system function $H(z)$ has a spectral null (i.e., a zero), at some frequency ω_0. Then the filter with system function $H_L(z)$ has periodically spaced nulls at $\omega_k = \omega_0 + 2\pi k/L$, $k = 0, 1, 2, \ldots,$ $L - 1$. As **an** illustration, Fig, 4.55 shows an FIR comb filter with M = 3 and L = 3. This FIR fitter can be viewed as an FIR filter of length 10, but only four of the 10 filter coefficients are nonzero.

Let us now return to the moving average filter with system function given by (4.5.33). Suppose that we replace z by z^L. Then the resulting comb filter has the system function

$$H_L(z) = \frac{1}{M+1}\,\frac{1 - z^{-L(M+1)}}{1 - z^{-L}} \tag{4.5.39}$$

and a frequency response

$$H_L(\omega) = \frac{1}{M+1}\,\frac{\sin[\omega L(M+1)/2]}{\sin(\omega L/2)}e^{-j\omega LM/2} \tag{4.5.40}$$

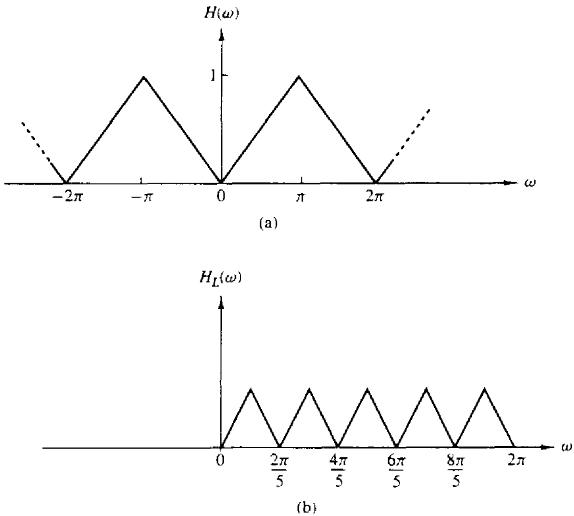


Figure 4.54 Comb filter with **frequency response** $H_L(\omega)$ **obtained** from $H(\omega)$.

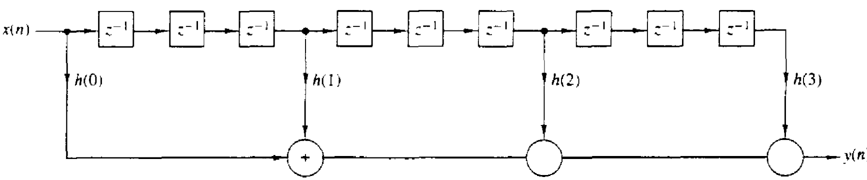


Figure 4.55 **Realization of an** FIR comb **filter having** $M = 3$ **and** $L = 3$.

This filter has zeros on the unit circle at

$$z_k = e^{j2\pi k/L(M+1)} \tag{4.5.41}$$

for all integer values of k except $k = 0, L, 2L, \ldots, ML$. Figure 4.56 illustrates $|H_L(\omega)|$ for $L = 5$ and $M = 10$.

The comb filter described by (4.5.39) finds application in the separation of solar and lunar spectral components in ionospheric measurements of electron concentration as described in the paper by Bernhardt et al. (1976). The solar period is $T_s = 24$ hours and results in a solar component of one cycle per day and its harmonics. The lunar period is $T_L = 24.84$ hours and provides spectral lines at 0.96618 cycle per day and its harmonics. Figure 4.57a shows a plot of the power density spectrum of the unfiltered ionospheric measurements of the electron con-

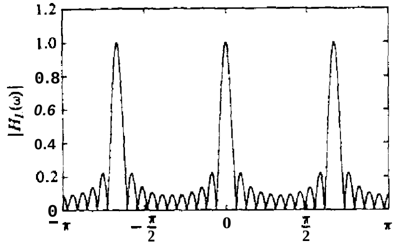


Figure 4.56 **Magnitude** response characteristic **for a** comb filter given by (4.5.40), with L = 3 and $M = 10$

centration. Note that the weak lunar spectral components are almost hidden by the strong solar spectral components.

The two sets of spectral components can be separated by the use of comb filters. If we wish to obtain the solar components. we can use a comb filler with a narrow passband at multiples of one cycle per day. This can be achieved by selecting L such that $F_s/L = 1$ cycle per day. where F_s is the corresponding sampling frequency. The result is a filter that has peaks in its frequency response at multiples of one cycle per day. By selecting M = 58. the filter will have nulls at multiples of $(F_s/L)/(M + 1) = 1/59$ cycle per day. These nulls are very close to the lunar components and result in good rejection. Figure 4.57(b) illustrates

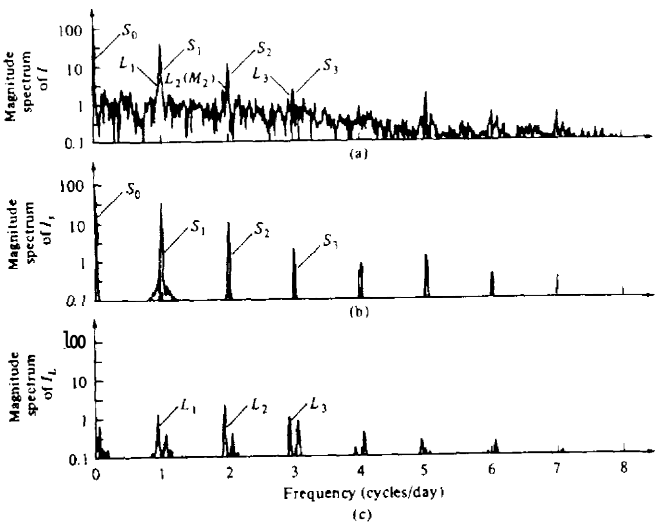


Figure 4.57 **(a) Spectrum of unfiltered electron content data; (b)** spectrum of output **of solar filter; (c) spectrum** of output of **lunar** filter. **[From paper** by **Bernhardt et al. (1976). Reprinted with permission of the American Geophysical Union,]**

the power spectral density of the output of the comb filter that isolates the solar components. A comb filter that rejects the solar components and passes the lunar components can be designed in a similar manner. Figure 4.57(c) illustrates the power spectral density at the output of such a lunar filter.

4.5.6 All-Pass Filters

An all-pass filter is defined as a system that has a constant magnitude response for all frequencies. that is.

$$|H(\omega)| = 1 \qquad 0 \le \omega \le \pi \tag{4.5.42}$$

The simplest example of an all-pass filter is a pure delay system with system function

$$H(z) = z^{-k}$$

This system passes all signals without modification except for a delay of k samples. This is a trivial all-pass system that has a linear phase response characteristic.

A more interesting all-pass filter is described by the system function

$$\begin{aligned} H(z) &= \frac{a_N + a_{N-1}z^{-1} + \cdots + a_1 z^{-N+1} + z^{-N}}{1 + a_1 z^{-1} + \cdots + a_N z^{-N}} \\ &= \frac{\sum_{k=0}^{N} a_k z^{-N+k}}{\sum_{k=0}^{N} a_k z^{-k}} \qquad a_0 = 1 \end{aligned} \tag{4.5.43}$$

where all the filter coefficients $\{a_k\}$ are real. If we define the polynomial $A(z)$ as

$$A(z) = \sum_{k=0}^{N} a_k z^{-k} \qquad a_0 = 1$$

then (4.5.43) can be expressed as

$$H(z) = z^{-N} \frac{A(z^{-1})}{A(z)} \tag{4.5.44}$$

Since

$$|H(\omega)|^2 = H(z)H(z^{-1})|_{z=e^{j\omega}} = 1$$

the system given by (4.5.44) is an all-pass system. Furthermore. if z_0 is a pole of $H(z)$. then $1/z_0$ is a zero of $H(z)$ (i.e., the poles and zeros are reciprocals of one another). Figure **4.58** illustrates typical pole–zero patterns for a single-pole, single-zero filter and a two-pole, two-zero filter. A plot of the phase characteristics of these filters is shown in Fig. **4.59** for a $= 0.6$ and $r = 0.9$. $\omega_0 = \pi/4$.

The most general form for the system function of an all-pass system with real coefficients, expressed in factored form in terms of poles and zeros. is

$$H_{\text{ap}}(z) = \prod_{k=1}^{N_R} \frac{z^{-1} - \alpha_k}{1 - \alpha_k z^{-1}} \prod_{k=1}^{N_c} \frac{(z^{-1} - \beta_k)(z^{-1} - \beta_k^*)}{(1 - \beta_k z^{-1})(1 - \beta_k^* z^{-1})} \tag{4.5.45}$$

where there are N_R real poles and zeros and N_c complex-conjugate pairs of poles and zeros. For causal and stable systems we require that $-1 < \alpha_k < 1$ and $|\beta_k| < 1$.

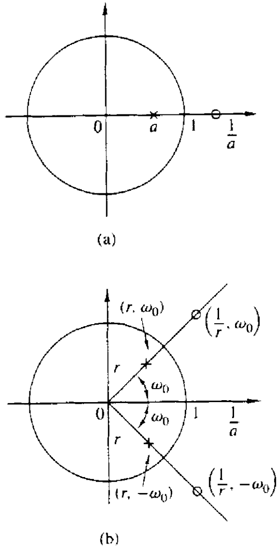


Figure 4.58 Pole–zero patterns of (a) a first-order and (b) a second-order all-pass filter.

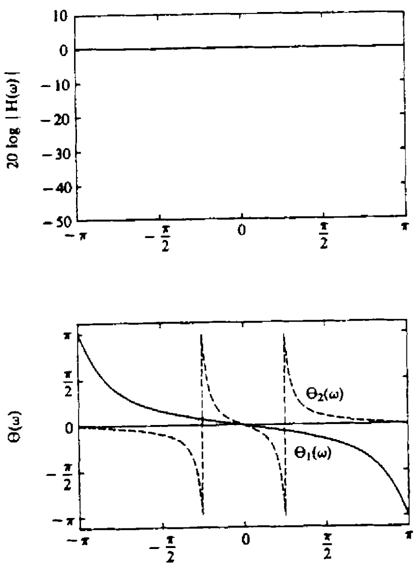


Figure 4.59 Frequency response characteristics of an all-pass filter with system functions (1) $H(z) = (0.6 + z^{-1})/(1 + 0.6z^{-1})$, (2) $H(z) = (r^2 - 2r\cos\omega_0 z^{-1} + z^{-2})/(1 - 2r\cos\omega_0 z^{-1} + r^2 z^{-2})$, $r = 0.9$,

Expressions for the phase response and group delay of all-pass systems can easily be obtained using the method described in Section 4.4.6. For a single pole–single zero all-pass system we have

$$H_{\text{ap}}(\omega) = \frac{e^{j\omega} - re^{-j\theta}}{1 - re^{j\theta}e^{-j\omega}}$$

Hence

$$\Theta_{\text{ap}}(\omega) = -\omega - 2\tan^{-1}\frac{r\sin(\omega - \theta)}{1 - r\cos(\omega - \theta)}$$

and

$$\tau_g(\omega) = -\frac{d\Theta_{\text{ap}}(\omega)}{d\omega} = \frac{1 - r^2}{1 + r^2 - 2r\cos(\omega - \theta)} \tag{4.5.46}$$

We note that for a causal and stable system. $r < 1$ and hence $\tau_g(\omega) \geq 0$. Since the group delay of a higher-order pole–zero system consists of a sum of positive terms as in (4.5.46), the group delay will always be positive.

All-pass filters find application as phase equalizers. When placed in cascade with a system that has an undesired phase response. a phase equalizer is designed to compensate for the poor phase characteristics of the system and therefore to produce an overall linear-phase response.

4.5.7 Digital Sinusoidal Oscillators

A digital sinusoidal oscillator can be viewed as a limiting form of a two-pole resonator for which the complex-conjugate poles lie on the unit circle. From our previous discussion of second-order systems, we recall that a system with system function

$$H(z) = \frac{b_0}{1 + a_1 z^{-1} + a_2 z^{-2}} \tag{4.5.47}$$

and parameters

$$a_1 = -2r\cos\omega_0 \quad \text{and} \quad a_2 = r^2 \tag{4.5.48}$$

has complex-conjugate poles at $p = re^{\pm j\omega_0}$, and a unit sample response

$$h(n) = \frac{b_0 r^n}{\sin\omega_0}\sin(n+1)\omega_0 u(n) \tag{4.5.49}$$

If the poles are placed on the unit circle ($r = 1$) and b_0 is set to $A\sin\omega_0$, then

$$h(n) = A\sin(n+1)\omega_0 u(n) \tag{4.5.50}$$

Thus the impulse response of the second-order system with complex-conjugate poles on the unit circle is a sinusoid and the system is called a digital sinusoidal oscillator or a ***digital sinusoidal*** generator. **A** digital sinusoidal generator is a basic component of a digital frequency synthesizer.

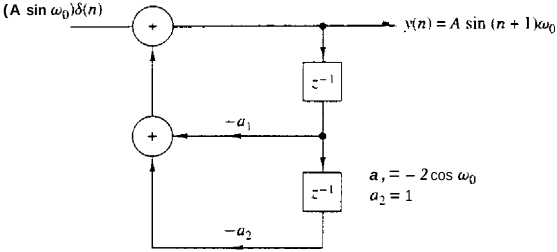


Figure 4.60 Digital sinusoidal generator

The block diagram representation of the system function given by (4.5.47) is illustrated in Fig. 4.60. The corresponding difference equation for this system is

$$y(n) = -a_1 y(n-1) - y(n-2) + b_0\delta(n) \tag{4.5.51}$$

where the parameters are $a_1 = -2\cos\omega_0$ and $b_0 = A\sin\omega_0$, and the initial conditions are $y(-1) = y(-2) = 0$. By iterating the difference equation in (4.5.51), we obtain

$$\begin{aligned}
y(0) &= A\sin\omega_0 \\
y(1) &= 2\cos\omega_0 y(0) = 2A\sin\omega_0\cos\omega_0 = A\sin 2\omega_0 \\
y(2) &= 2\cos\omega_0 y(1) - y(0) \\
&= 2A\cos\omega_0\sin 2\omega_0 - A\sin\omega_0 \\
&= A(4\cos^2\omega_0 - 1)\sin\omega_0 \\
&= 3A\sin\omega_0 - 4\sin^3\omega_0 = A\sin 3\omega_0
\end{aligned}$$

and so forth. We note that the application of the impulse at $n = 0$ serves the purpose of beginning the sinusoidal oscillation. Thereafter, the oscillation is self-sustaining because the system has no damping (i.e., $r = 1$).

It is interesting to note that the sinusoidal oscillation obtained from the system in (4.5.51) can also be obtained by setting the input to zero and setting the initial conditions to $y(-1) = 0$, $y(-2) = -A\sin\omega_0$. Thus the zero-input response to the second-order system described by the homogeneous difference equation

$$y(n) = -a_1 y(n-1) - y(n-2) \tag{4.5.52}$$

with initial conditions $y(-1) = 0$ and $y(-2) = -A\sin\omega_0$, is exactly the same as the response of (4.5.51) to an impulse excitation. In fact, the difference equation in (4.5.52) can be obtained directly from the trigonometric identity

$$\sin\alpha + \sin\beta = 2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2} \tag{4.5.53}$$

where, by definition, $\alpha = (n+1)\omega_0$, $\beta = (n-1)\omega_0$, and $y(n) = \sin(n+1)\omega_0$.

In some practical applications involving modulation of two sinusoidal carrier signals in phase quadrature, there is a need to generate the sinusoids $A \sin \omega_0 n$ and $A \cos \omega_0 n$. These signals can be generated from the so-called coupled-form *oscillator*, which can be obtained from the trigonometric formulas

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

where, by definition, $\alpha = n\omega_0$, $\beta = \omega_0$, and

$$y_c(n) = \cos n\omega_0 u(n) \tag{4.5.54}$$

$$y_s(n) = \sin n\omega_0 u(n) \tag{4.5.55}$$

Thus we obtain the two coupled difference equations

$$y_c(n) = (\cos \omega_0) y_c(n-1) - (\sin \omega_0) y_s(n-1) \tag{4.5.56}$$

$$y_s(n) = (\sin \omega_0) y_c(n-1) + (\cos \omega_0) y_s(n-1) \tag{4.5.57}$$

which can also be expressed in matrix form as

$$\begin{bmatrix} y_c(n) \\ y_s(n) \end{bmatrix} = \begin{bmatrix} \cos \omega_0 & -\sin \omega_0 \\ \sin \omega_0 & \cos \omega_0 \end{bmatrix} \begin{bmatrix} y_c(n-1) \\ y_s(n-1) \end{bmatrix} \tag{4.5.58}$$

The structure for the realization of the coupled-form oscillator is illustrated in Fig. 4.61. We note that this is a two-output system which is not driven by any input, but which requires the initial conditions $y_c(-1) = A \cos \omega_0$ and $y_s(-1) = -A \sin \omega_0$ in order to begin its self-sustaining oscillations.

Finally, it is interesting to note that (4.5.58) corresponds to vector rotation in the two-dimensional coordinate system with coordinates $y_c(n)$ and $y_s(n)$. As a consequence, the coupled-form oscillator can also be implemented by use of the so-called CORDIC algorithm [see the book by Kung et al. (1985)].

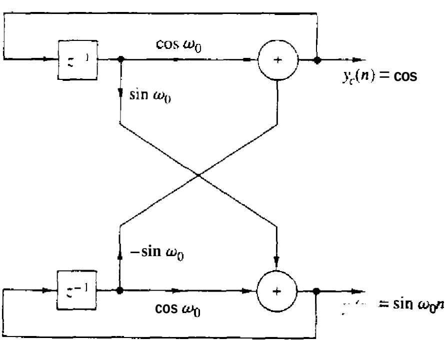


Figure 4.61 Realization of the coupled-form oscillator.

4.6 INVERSE SYSTEMS AND DECONVOLUTION

As we have seen. a linear time-invariant system takes an input signal $x(n)$ and produces an output signal $y(n)$. which is the convolution of $x(n)$ with the unit sample response $h(n)$ of the system. In many practical applications we are given an output signal from a system whose characteristics are unknown and we are asked to determine the input signal. For example. in the transmission of digital information at high data rates over telephone channels. it is well known that the channel distorts the signal and causes intersymbol interference among the data symbols. The intersymbol interference may cause errors when we attempt to recover the data. In such a case the problem is to design a corrective systcm which. when cascaded with the channel. produces an output that. in some sense. corrects for the distortion caused by the channel. and thus yields a replica of the desired transmitted signal. In digital communications such a corrective system is called an *equalizer*. In the general context of linear systems theory. however. we call the corrective svstcm an *inverse system*. because the corrective system has a frequency response which is basically the reciprocal of the frequency response of the system that caused the distortion. Furthermore. since the distortive system yields an output $y(n)$ that is the convolution of the input $x(n)$ with the impulse response $h(n)$. the inverse system operation that takes $y(n)$ and produces $x(n)$ is called *deconvolution*.

If the characteristics of the distortive system are unknown. it is often necessary. when possible. to excite the system with a known signal. observe the output, compare it with the input. and in some manner, determine the characteristics of the system. For example, in the digital communication problem just described, where the frequency response of the channel is unknown. the measurement of the channel frequency response can be accomplished by transmitting a set of equal amplitude sinusoids, at different frequencies with a specified set of phases. within the frequency band of the channel. The channel will attenuate and phase shift each of the sinusoids. By comparing the received signal with the transmitted signal. the receiver obtains a measurement of the channel frequency response which can be used to design the inverse system. The process of determining the characteristics of the unknown system, either $h(n)$ or $H(\omega)$. *by* a set of measurements performed on the system is called *system identification*.

The term "deconvolution" is often used in seismic signal processing. and more generally, in geophysics to describe the operation of separating the input signal from the characteristics of the system which we wish to measure. The deconvolution operation is actually intended to identify the characteristics of the system, which in this case, is the earth. and can also be viewed as a system identification problem. The "inverse system," in this case. has a frequency response that is the reciprocal of the input signal spectrum that has been used to excite the system.

4.6.1 Invertibility of Linear Time-Invariant Systems

A system is said to be ***invertible*** if there is a one-to-one correspondence between its input and output signals. This definition implies that if we know the output sequence $y(n)$, $-\infty < n < \infty$, of an invertible system $\mathcal{T}$, we can uniquely determine its input $x(n)$, $-\infty < n < \infty$. The ***inverse system*** with input $y(n)$ and output $x(n)$ is denoted by $\mathcal{T}^{-1}$. Clearly, the cascade connection of a system and its inverse is equivalent to the identity system, since

$$w(n) = \mathcal{T}^{-1}[y(n)] = \mathcal{T}^{-1}\{\mathcal{T}[x(n)]\} = x(n) \tag{4.6.1}$$

as illustrated in Fig. 4.62. For example, the systems defined by the input–output relations $y(n) = ax(n)$ and $y(n) = x(n-5)$ are invertible, whereas the input–output relations $y(n) = x^2(n)$ and $y(n) = 0$ represent noninvertible systems.

As indicated above, inverse systems are important in many practical applications, including geophysics and digital communications. Let us begin by considering the problem of determining the inverse of a given system. We limit our discussion to the class of linear time-invariant discrete-time systems.

Now, suppose that the linear time-invariant system $\mathcal{T}$ has an impulse response $h(n)$ and let $h_I(n)$ denote the impulse response of the inverse system $\mathcal{T}^{-1}$. Then (4.6.1) is equivalent to the convolution equation

$$w(n) = h_I(n) * h(n) * x(n) = x(n) \tag{4.6.2}$$

But (4.6.2) implies that

$$h(n) * h_I(n) = \delta(n) \tag{4.6.3}$$

The convolution equation in (4.6.3) can be used to solve for $h_I(n)$ for a given $h(n)$. However, the solution of (4.6.3) in the time domain is usually difficult. A simpler approach is to transform (4.6.3) into the z-domain and solve for $\mathcal{T}^{-1}$. Thus in the z-transform domain, (4.6.3) becomes

$$H(z)H_I(z) = 1$$

and therefore the system function for the inverse system is

$$H_I(z) = \frac{1}{H(z)} \tag{4.6.4}$$

If $H(z)$ has a rational system function

$$H(z) = \frac{B(z)}{A(z)} \tag{4.6.5}$$

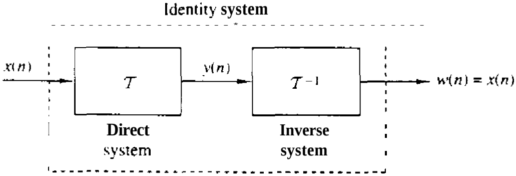


Figure 4.62 System $\mathcal{T}$ in cascade with its inverse $\mathcal{T}^{-1}$.

then

$$H_I(z) = \frac{A(z)}{B(z)} \tag{4.6.6}$$

Thus the zeros of $H(z)$ become the poles of the inverse system, and vice versa. Furthennore, if $H(z)$ is an FIR system, then $H_I(z)$ is an all-pole system. or if $H(z)$ is an all-pole system, then $H_I(z)$ is an FIR system.

Example 4.6.1

Determine the inverse of the system with impulse response

$$h(n) = (\tfrac{1}{2})^n u(n)$$

Solution The system function corresponding to $h(n)$ is

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} \qquad \textbf{ROC: } |z| > \tfrac{1}{2}$$

This system is both causal and stable. Since $H(z)$ is an all-pole system, its inverse is FIR and is given by the system function

$$H_I(z) = 1 - \tfrac{1}{2}z^{-1}$$

Hence its impulse response is

$$h_I(n) = \delta(n) - \tfrac{1}{2}\delta(n-1)$$

Example 4.6.2

Determine the inverse of the system with impulse response

$$h(n) = \delta(n) - \tfrac{1}{2}\delta(n-1)$$

Solution This is an FIR system **and** its system function is

$$H(z) = 1 - \tfrac{1}{2}z^{-1} \qquad \text{ROC: } |z| > 0$$

The inverse system has the system function

$$H_I(z) = \frac{1}{H(z)} = \frac{1}{1 - \frac{1}{2}z^{-1}} = \frac{z}{z - \frac{1}{2}}$$

Thus $H_I(z)$ has a zero at the origin and a pole at $z = \frac{1}{2}$. In this case there are **two** possible regions of convergence and hence ***two*** possible inverse systems, as illustrated in Fig. 4.63. If we take the ROC of $H_I(z)$ as $|z| > \frac{1}{2}$, the inverse transform yields

$$h_I(n) = (\tfrac{1}{2})^n u(n)$$

which is the impulse response of a causal and stable system. On the other hand, if the ROC is assumed to be $|z| < \frac{1}{2}$, the inverse system has an impulse response

$$h_I(n) = -\left(\frac{1}{2}\right)^n u(-n-1)$$

In this case the inverse system is anticausal and unstable.

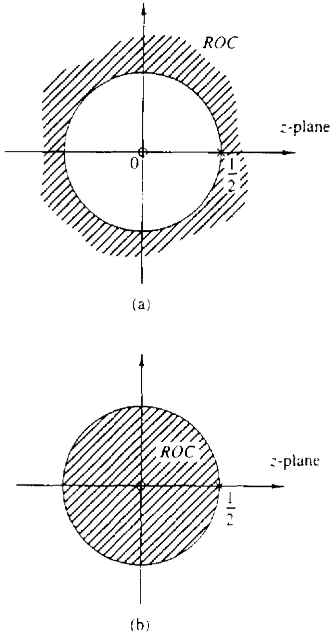


Figure 4.63 Two possible regions of convergence for $H(z) = z/(z - \frac{1}{2})$.

We observe that (4.6.3) cannot be solved uniquely by using (4.6.6) unless we specify the region of convergence for the system function of the inverse system.

In some practical applications the impulse response $h(n)$ does not possess a z-transform that can be expressed in closed form. As an alternative we may solve (4.6.3) directly using a digital computer. Since (4.6.3) does not. in general. possess a unique solution, we assume that the system and its inverse are causal. Then (4.6.3) simplifies to the equation

$$\sum_{k=0}^{n} h(k)h_I(n-k) = \delta(n) \tag{4.6.7}$$

By assumption. $h_I(n) = 0$ for $n < 0$. For $n = 0$ we obtain

$$h_I(0) = 1/h(0) \tag{4.6.8}$$

The values of $h_I(n)$ for $n \geq 1$ can be obtained recursively from the equation

$$h_I(n) = -\sum_{k=1}^{n} \frac{h(k)h_I(n-k)}{h(0)} \qquad n \geq 1 \tag{4.6.9}$$

This recursive relation can easily be programmed on a digital computer.

There are two problems associated with *(4.6.9)*. First, the method does not work if $h(0) = 0$. However. this problem can easily be remedied by introducing an appropriate delay in the right-hand side of (4.6.7), that is, by replacing $\delta(n)$ by $S(n-m)$, where $m = 1$ if $h(0) = 0$ and $h(1) \neq 0$, and so on. Second, the recursion in (4.6.9) gives rise to round-off errors which grow with n and, as a result, the numerical accuracy of $h(n)$ deteriorates for large n.

Example 4.6.3

Determine the causal inverse of the FIR system with impulse response

$$h(n) = \delta(n) - \alpha\delta(n-1)$$

Solution Since $h(0) = 1$. $h(1) = -\alpha$, and $h(n) = 0$ for $n \geq \alpha$, we have

$$h_I(0) = 1/h(0) = 1$$

and

$$h_I(n) = \alpha h_I(n-1) \qquad n \geq 1$$

Consequently.

$$h_I(1) = \alpha, \quad h_I(2) = \alpha^2, \quad \ldots, \quad h_I(n) = \alpha^n$$

which corresponds to a causal IIR system as expected.

4.6.2 Minimum-Phase, Maximum-Phase, and Mixed-Phase Systems

The invertibility of a linear time-invariant system is intimately related to the characteristics of the phase spectral function of the system. To illustrate this point, let us consider two FIR systems, characterized by the system functions

$$H_1(z) = 1 + \tfrac{1}{2}z^{-1} = z^{-1}(z + \tfrac{1}{2}) \tag{4.6.10}$$

$$H_2(z) = \tfrac{1}{2} + z^{-1} = z^{-1}(\tfrac{1}{2}z + 1) \tag{4.6.11}$$

The system in (4.6.10) has a zero at $z = -\frac{1}{2}$ and an impulse response $h(0) = 1$, $h(1) = 1/2$. The system in *(4.6.11)* has a zero at $z = -2$ and an impulse response $h(0) = 1/2$, $h(1) = 1$, which is the reverse of the system in *(4.6.10)*. This is due to the reciprocal relationship between the zeros of $H_1(z)$ and $H_2(z)$.

In the frequency domain, the two systems are characterized by their frequency response functions, which can be expressed as

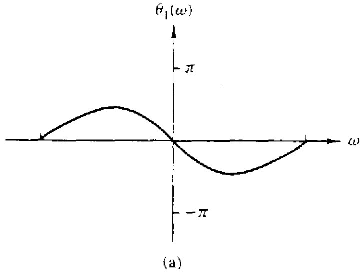
$$|H_1(\omega)| = |H_2(\omega)| = \sqrt{\tfrac{5}{4} + \cos\omega} \tag{4.6.12}$$

and

$$\Theta_1(\omega) = -\omega + \tan^{-1}\frac{\sin\omega}{\frac{1}{2} + \cos\omega} \tag{4.6.13}$$

$$\Theta_2(\omega) = -\omega + \tan^{-1}\frac{\sin\omega}{2 + \cos\omega} \tag{4.6.14}$$

The magnitude characteristics **for** the two systems are identical because the zeros of $H_1(z)$ and $H_2(z)$ are reciprocals.



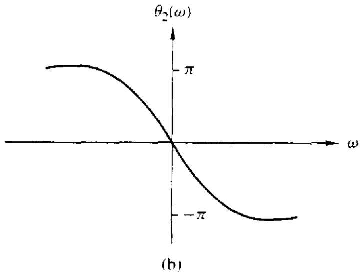


Figure 4.64 Phase response characteristics for the systems in (4.6.10) and (4.6.11).

The graphs of $\Theta_1(\omega)$ and $\Theta_2(\omega)$ are illustrated in Fig. 4.64. We observe that the phase characteristic $\Theta_1(\omega)$ for the first system begins at zero phase at the frequency $w = 0$ and terminates at zero phase at the frequency $\omega = \pi$. Hence the net phase change, $\Theta_1(\pi) - \Theta_1(0)$ is zero. On the other hand, the phase characteristic for the system with the zero outside the unit circle undergoes a net phase change $\Theta_2(\pi) - \Theta_2(0) = \pi$ radians. As a consequence of these different phase characteristics, we call the first system a ***minimum-phase** system* and the second system is called a maximum-phase system.

These definitions are easily extended to an FIR system of arbitrary length. To be specific, an FIR system of length M + 1 has M zeros. Its frequency response can be expressed as

$$H(\omega) = b_0(1 - z_1e^{-j\omega})(1 - z_2e^{-j\omega}) \cdots (1 - z_Me^{-j\omega}) \tag{4.6.15}$$

where $\{z_i\}$ denote the zeros and b_0 is an arbitrary constant. When all the zeros are inside the unit circle. each term in the product of (4.6.15), corresponding to a real-valued zero, will undergo a net phase change of zero between $\omega = 0$ and $\omega = \pi$. Also, each pair of complex- conjugate factors in $H(\omega)$ will undergo a net phase change of zero. Therefore,

$$\measuredangle H(\pi) - \measuredangle H(0) = 0 \tag{4.6.16}$$

and hence the system is called a minimum-phase system. On the other hand, when all the zeros are outside the unit circle, a real-valued zero will contribute a net phase change of π radians as the frequency varies from $\omega = 0$ to $\omega = \pi$, and each pair of complex-conjugate zeros will contribute a net phase change of 2π radians over the same range of ω. Therefore,

$$\measuredangle H(\pi) - \measuredangle H(0) = M\pi \tag{4.6.17}$$

which is the largest possible phase change for an FIR system with M zeros. Hence the system is called maximum phase. It follows from the discussion above that

$$\measuredangle H_{\text{max}}(\pi) \geq \measuredangle H_{\text{min}}(\pi) \tag{4.6.18}$$

If the FIR system with M zeros has some of its zeros inside the unit circle and the remaining zeros outside the unit circle, it is called a *mixed-phase system* or a *nonminimum-phase system*.

Since the derivative of the phase characteristic of the system is a measure of the time delay that signal frequency components undergo in passing through the system, a minimum-phase characteristic implies a minimum delay function, while a maximum-phase characteristic implies that the delay characteristic is also maximum.

Now suppose that we have an FIR system with real coefficients. Then the magnitude square value of its frequency response is

$$|H(\omega)|^2 = H(z)H(z^{-1})|_{z=e^{j\omega}} \tag{4.6.19}$$

This relationship implies that if we replace a zero z_k of the system by its inverse $1/z_k$, the magnitude characteristic of the system does not change. Thus if we reflect a zero z_k that is inside the unit circle into a zero $1/z_k$ outside the unit circle, we see that the magnitude characteristic of the frequency response is invariant to such a change.

It is apparent from this discussion that if $|H(\omega)|^2$ is the magnitude square frequency response of an FIR system having M zeros, there are 2^M possible configurations for the M zeros, some of which are inside the unit circle and the remaining are outside the unit circle. Clearly, one configuration has all the zeros inside the unit circle, which corresponds to the minimum-phase system. A second configuration has all the zeros outside the unit circle, which corresponds to the maximum-phase system. The remaining $2^M - 2$ configurations correspond to mixed-phase systems. However, not all $2^M - 2$ mixed-phase configurations necessarily correspond to FIR systems with real-valued coefficients. Specifically, any pair of complex-conjugate zeros result in only two possible configurations, whereas a pair of real-valued zeros yield four possible configurations.

Example 4.6.4

Determine the zeros for the following FIR systems and indicate whether the system is minimum phase, maximum phase, or mixed phase.

$$H_1(z) = 6 + z^{-1} - z^{-2}$$
$$H_2(z) = 1 - z^{-1} - 6z^{-2}$$

$$H_3(z) = 1 - \tfrac{5}{2}z^{-1} - \tfrac{3}{2}z^{-2}$$
$$H_4(z) = 1 + \tfrac{5}{3}z^{-1} - \tfrac{2}{3}z^{-2}$$

Solution By factoring the system functions we find the zeros for the four systems are

$$H_1(z) \longrightarrow z_{1,2} = -\tfrac{1}{2}, \tfrac{1}{3} \longrightarrow \text{minimum phase}$$
$$H_2(z) \longrightarrow z_{1,2} = -2, 3 \longrightarrow \text{maximum phase}$$
$$H_3(z) \longrightarrow z_{1,2} = -\tfrac{1}{2}, 3 \longrightarrow \text{mixed phase}$$
$$H_4(z) \longrightarrow z_{1,2} = -2, \tfrac{1}{3} \longrightarrow \text{mixed phase}$$

Since the zeros of the four systems are reciprocals of one another, it follows that all lour systems have identical magnitude frequency response characteristics hut different phase characteristics.

The minimum-phase property of FIR systems carries over to IIR systems that have rational system functions. Specifically, an IIR system with system function

$$H(z) = \frac{B(z)}{A(z)} \tag{4.6.20}$$

is called *minimum phase* if all its poles and zeros are inside the unit circle. For a stable and causal system [all roots of $A(z)$ fall inside the unit circle] the svstem is called *maximum phase* if all the zeros are outside the unit circle. and *mixed* phase if some. but not all. of the zeros are outside the unit circle.

This discussion brings us to an important point that should be emphasized. That is. a *stable* pole–zero system that is minimum phase has a stable inverse which is also minimum phase. The inverse system has the system function

$$H^{-1}(z) = \frac{A(z)}{B(z)} \tag{4.6.21}$$

Hence the minimum-phase property of $H(z)$ ensures the stability of the inverse system $H^{-1}(z)$ and the stability of $H(z)$ implies the minimum-phase property of $H^{-1}(z)$. Mixed-phase systems and maximum-phase systems result in unstable inverse systems.

Decomposition of nonminimum-phase pole-zero systems. Any nonminimum-phase pole–zero system can be expressed as

$$H(z) = H_{\min}(z)H_{\text{ap}}(z) \tag{4.6.22}$$

where $H_{\min}(z)$ is a minimum-phase system and $H_{\text{ap}}(z)$ is an all-pass system. We demonstrate the validity of this assertion for the class of causal and stable systems with a rational system function $H(z) = B(z)/A(z)$. In general, if $B(z)$ has one or more roots outside the unit circle, we factor $B(z)$ into the product $B_1(z)B_2(z)$, where $B_1(z)$ has all its roots inside the unit circle and $B_2(z)$ has all its roots outside the unit circle. Then $B_2(z^{-1})$ has all its roots inside the unit circle. We define the minimum-phase system

$$H_{\min}(z) = \frac{B_1(z)B_2(z^{-1})}{A(z)}$$

and the all-pass system

$$H_{\text{ap}}(z) = \frac{B_2(z)}{B_2(z^{-1})}$$

Thus $H(z) = H_{\min}(z)H_{\text{ap}}(z)$. Note that $H_{\text{ap}}(z)$ is a stable, all-pass. maximum-phase system.

Group delay of nonminimum-phase system. Based on the decomposition of **a** nonminirnurn-phase system given by (4.6.22), we can express the group delay of $H(z)$ as

$$\tau_g(\omega) = \tau_g^{\min}(\omega) + \tau_g^{\text{ap}}(\omega) \tag{4.6.23}$$

Since $\tau_g^{ap}(\omega) \geq 0$ for $0 \leq \omega \leq \pi$, it follows that $\tau_g(\omega) \geq \tau_g^{\min}(\omega)$, $0 \leq \omega \leq \pi$. From *(4.6.23)* we conclude that among all pole–zero systems having the same magnitude response, the minimum-phase system has the smallest group delay.

Partial energy of nonminimum-phase system. The partial energy of a causal system with impulse response $h(n)$ is defined as

$$E(n) = \sum_{k=0}^{n} |h(k)|^2 \tag{4.6.24}$$

It can be shown that among all systems having the same magnitude response and the same total energy $E(\infty)$, the minimum-phase system has the largest partial energy [i.e., $E_{\min}(n) \geq E(n)$, where $E_{\min}(n)$ is the partial energy of the minimum-phase system].

4.6.3 System Identification and Deconvolution

Suppose that we excite an unknown linear time-invariant system with an input sequence $x(n)$ and we observe the output sequence $y(n)$. From the output sequence we wish to determine the impulse response of the unknown system. This is a problem in *system identification,* which can be solved by *deconvolution.* Thus **we** have

$$\begin{aligned} y(n) &= h(n) * x(n) \\ &= \sum_{k=-\infty}^{\infty} h(k)x(n-k) \end{aligned} \tag{4.6.25}$$

An analytical solution of the deconvolution problem **can** be obtained by working with the z-transform of *(4.6.25).* In the z-transform domain we have

$$Y(z) = H(z)X(z)$$

and hence

$$H(z) = \frac{Y(z)}{X(z)} \tag{4.6.26}$$

$X(z)$ and $Y(z)$ are the z-transforms of the available input signal $x(n)$ and the observed output signal $y(n)$, respectively. This approach is appropriate only when there are closed-form expressions for $X(z)$ and $Y(z)$.

Example 4.6.5

A causal system produces the output sequence

$$y(n) = \begin{cases} 1, & n = 0 \\ \frac{7}{10}, & n = 1 \\ 0, & \text{otherwise} \end{cases}$$

when excited by the input sequence

$$x(n) = \begin{cases} 1, & n = 0 \\ -\frac{7}{10}, & n = 1 \\ \frac{1}{10}, & n = 2 \\ 0, & \text{otherwise} \end{cases}$$

Determine its impulse response and its input–output equation.

Solution The system function is easily determined by taking the z-transforms of $x(n)$ and $y(n)$. Thus we have

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + \frac{7}{10}z^{-1}}{1 - \frac{7}{10}z^{-1} + \frac{1}{10}z^{-2}}$$

$$= \frac{1 + \frac{7}{10}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{5}z^{-1})}$$

Since the system is causal, its ROC is $|z| > \frac{1}{2}$. The system is also stable since its poles lie inside the unit circle.

The input–output difference equation for the system is

$$y(n) = \frac{7}{10}y(n-1) - \frac{1}{10}y(n-2) + x(n) + \frac{7}{10}x(n-1)$$

Its impulse response is determined by performing a partial-fraction expansion of $H(z)$ and inverse transforming the result. This computation yields

$$h(n) = [4(\tfrac{1}{2})^n - 3(\tfrac{1}{5})^n]u(n)$$

We observe that (4.6.26) determines the unknown system uniquely if it is known that the system is causal. However, the example above is artificial, since the system response $\{y(n)\}$ is very likely to be infinite in duration. Consequently, this approach is usually impractical.

As an alternative, we can deal directly with the time-domain expression given by (4.6.25). If the system is causal, we have

$$y(n) = \sum h(k)x(n-k) \qquad n \geq 0$$

and hence

$$
\begin{aligned}
h(0) &= \frac{y(0)}{x(0)} \\
h(n) &= \frac{y(n) - \sum_{k=0}^{n-1} h(k)x(n-k)}{x(0)} \qquad n \geq 1
\end{aligned}
\tag{4.6.27}
$$

This recursive solution requires that $x(0) \neq 0$. However, *we* note again that when $\{h(n)\}$ has infinite duration, this approach may not be practical unless we truncate the recursive solution at same stage [i.e., truncate $\{h(n)\}$].

Another method for identifying an unknown system is based on a crosscorrelation technique. Recall that the input-output crosscorrelation function derived in Section *2.6.5* is given as

$$r_{yx}(m) = \sum_{k=0}^{\infty} h(k) r_{xx}(m-k) = h(n) * r_{xx}(m) \tag{4.6.28}$$

where $r_{yx}(m)$ is the crosscorrelation sequence of the input $\{x(n)\}$ to the system with the output $\{y(n)\}$ of the system, and $r_{xx}(m)$ is the autocorrelation sequence of the input signal. In the frequency domain, the corresponding relationship is

$$S_{yx}(\omega) = H(\omega) S_{xx}(\omega) = H(\omega) |X(\omega)|^2$$

Hence

$$H(\omega) = \frac{S_{yx}(\omega)}{S_{xx}(\omega)} = \frac{S_{yx}(\omega)}{|X(\omega)|^2} \tag{4.6.29}$$

These relations suggest that the impulse response $\{h(n)\}$ or the frequency response of an unknown system can be determined (measured) by crosscorrelating the input sequence $\{x(n)\}$ with the output sequence $\{y(n)\}$, and then solving the deconvolution problem in (4.6.28) by means of the recursive equation in (4.6.27). Alternatively, we could simply compute the Fourier transform of (4.6.28) and determine the frequency response given by *(4.6.29).* Furthermore, if we select the input sequence *(* $x(n)\}$ such that its autocorrelation sequence *(r,,* $(n)\}$, is *a* unit sample sequence, or equivalently, that its spectrum is flat (constant) over the passband of $H(\omega)$, the values of the impulse response $\{h(n)\}$ are simply equal to the values of the crosscorrelation sequence *(r,,* $(n)\}$.

In general, the crosscorrelation method described above is an effective and practical method for system identification. Another practical approach based on least-squares optimization is described in Chapter 8.

4.6.4 Homomorphic Deconvolution

The complex cepstrum, introduced in Section *4.2.7,* is *a* useful tool for performing deconvolution in some applications such as seismic signal processing. To describe this method, let us suppose that $\{y(n)\}$ is the output sequence of a linear time-invariant system which is excited by the input sequence $\{x(n)\}$. Then

$$Y(z) = X(z)H(z) \tag{4.6.30}$$

where $H(z)$ is the system function. The logarithm of $Y(z)$ is

$$\begin{aligned} C_y(z) &= \ln Y(z) \\ &= \ln X(z) + \ln H(z) \\ &= C_x(z) + C_h(z) \end{aligned} \tag{4.6.31}$$

Consequently, the complex cepstrum of the output sequence $\{y(n)\}$ is expressed as the sum of the cepstrum of $\{x(n)\}$ and $\{h(n)\}$, that is,

$$c_y(n) = c_x(n) + c_h(n) \tag{4.6.32}$$

Thus we observe that convolution of the two sequences in the time domain corresponds to the summation of the cepstrum sequences in the cepsrral domain. The system for performing these transformations is called a *homormorphic system* and is illustrated in Fig. **4.65.**

In some applications, such as seismic signal processing and speech signal processing. the characteristics of the cepstral sequences $\{c_x(n)\}$ and $\{c_h(n)\}$ are sufficiently different so that they can be separated in the cepstral domain. Specifically, suppose that $\{c_h(n)\}$ has its main components (main energy) in the vicinity of small values of n, whereas $\{c_x(n)\}$ has its components concentrated at large values of n. We may say rhat $\{c_h(n)\}$ is "lowpass" and $\{c_x(n)\}$ is "highpass." *We* can then separate $\{c_h(n)\}$ from $\{c_x(n)\}$ using appropriate "lowpass" and "highpass" windows. as illustrated in Fig. **4.66.** Thus

$$\hat{c}_h(n) = c_y(n)w_{\text{lp}}(n) \tag{4.6.33}$$

and

$$\hat{c}_x(n) = c_y(n)w_{\text{hp}}(n) \tag{4.6.34}$$

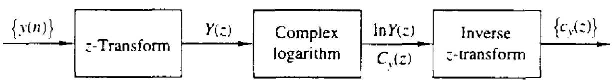


Figure 4.65 Homomorphic svstern for obtaining the cepstrum $\{c_y(n)\}$ of the sequence $\{y(n)\}$.

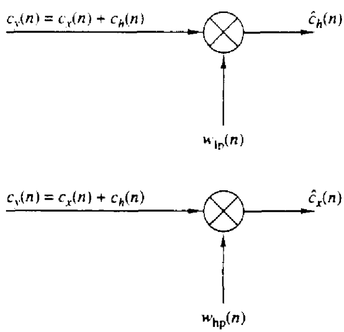


Figure 4.66 Separating the two cepstral components by "lowpass" and "highpass" windows.

where

$$w_{lp}(n) = \begin{cases} 1, & |n| \le N_1 \\ 0, & \text{otherwise} \end{cases} \tag{4.6.35}$$

$$w_{hp}(n) = \begin{cases} 0, & |n| \le N_1 \\ 1, & |n| > N_1 \end{cases} \tag{4.6.36}$$

Once we have separated the cepstrum sequences $\{\hat{c}_h(n)\}$ and $\{\hat{c}_x(n)\}$ by windowing, the sequences $\{\hat{x}(n)\}$ and $\{\hat{h}(n)\}$ are obtained by passing $\{\hat{c}_h(n)\}$ and $\{\hat{c}_x(n)\}$ through the inverse hornomorphic system. shown in Fig. 4.67.

In practice. a digital computer would be used to compute the cepstrum of the sequence $\{y(n)\}$, to perform the windowing functions, and to implement the inverse homomorphic system shown in Fig. 4.67. In place of the z-transform and inverse z-transform, we would substitute a special form of the Fourier transform and its inverse. This special form, called the discrete Fourier transform, is described in Chapter 5.

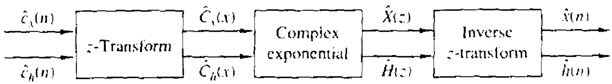


Figure 4.67 Inverse homomorphic system for recovering the sequences $\{x(n)\}$ and $\{h(n)\}$ from the corresponding cepstra.

4.7 SUMMARY AND REFERENCES

The Fourier series and the Fourier transform are the mathematical tools lor analyzing the characteristics of signals in the frequency domain. The Fourier series is appropriate for representing a periodic signal as a weighted sum of harmonically related sinusoidal components. where the weighting coefficients represent the strengths of each of the harmonics, and the magnitude squared of each weighting coefficient represents the power of the corresponding harmonic. As we have indicated, the Fourier series is one of many possible orthogonal series expansions for a periodic signal. Its importance stems from the characteristic behavior of LTI systems, as we shall see in Chapter 5.

The Fourier transform is appropriate for representing the spectral characteristics of aperiodic signals with finite energy. The important properties of the Fourier transform were also presented in this chapter.

There are many excellent texts on Fourier series and Fourier transforms. For reference, we include the texts by Bracewell (1978). Davis (1963), Dym and McKean (1972). and Papoulis (1962).

In this chapter we also considered the frequency-domain characteristics of LTI systems. We showed that an LTI system is characterized in the frequency domain by its frequency response function $H(\omega)$, which is the Fourier transform of the impulse response of the system. We also observed that the frequency response function determines the effect of the system on any input signal. In fact, by transforming the input signal into the frequency domain, we observed that it is a simple matter to determine the effect of the system on the signal and to determine the system output. When viewed in the frequency domain, an LTI system performs spectral shaping or spectral filtering on the input signal.

The design of some simple IIR filters was also considered in this chapter from the viewpoint of pole–zero placement. By means of this method. we were able to design simple digital resonators, notch filters, comb filters. all-pass filters. and digital sinusoidal generators. The design of more complex IIR filters is treated in detail in Chapter 8. which also includes several references. Digital sinusoidal generators find use in frequency synthesis applications. A comprehensive treatment of frequency synthesis techniques is given in the text edited by Gorski-Popiel (1975).

Finally. we characterized LTI systems as either minimum-phase, maximum-phase, or mixed-phase. depending on the position of their poles and zeros in the frequency domain. Using these basic characteristics of LTI systems. we considered practical problems in inverse filtering. deconvolution. and system identification. We concluded with the description of a deconvolution method based on cepstral analysis of the output signal from a linear system.

A vast amount of technical literature exists on the topics of inverse filtering. deconvolution, and system identification. In the context of communications, system identification. and inverse filtering as they relate to channel equalization are rreated in the book by Proakis (199.5). Deconvolution techniques are widely used in seismic signal processing. For reference. we suggest the papers by Wood and Treitel (1975). Peacock and Treitel (1969). and the books by Robinson and Treitel (1978. 1980). Homomorphic deconvolution and its applications to speech processing is treated in the book by Oppenheim and Schafer (1989).

PROBLEMS

4.1 Consider the full-wave rectified sinusoid in Fig. **P4.1.**

(a) Determine its spectrum $X_a(F)$.

(b) Compute the power of the signal.

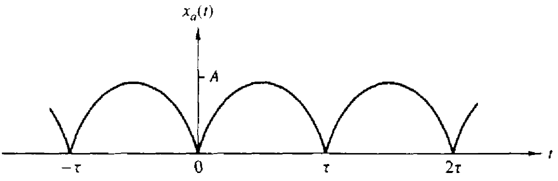


Figure P4.1

(c) Plot the power spectral density.
(d) Check the validity of Parseval's relation for this signal.

4.1 Compute and sketch the magnitude and phase spectra for the following signals ($a > 0$).

(a) $x_a(t) = \begin{cases} Ae^{-at}, & t \geq 0 \\ 0, & t < 0 \end{cases}$

(b) $x_a(t) = Ae^{-a|t|}$

4.3 Consider the signal

$$x(t) = \begin{cases} 1 - |t|/\tau, & |t| \leq \tau \\ 0, & \text{elsewhere} \end{cases}$$

(a) Determine and sketch its magnitude and phase spectra, $|X_a(F)|$ and $\measuredangle X_a(F)$, respectively.
(b) Create a periodic signal $x_p(t)$ with fundamental period $T_p \geq 2\tau$, so that $x(t) = x_p(t)$ for $|t| < T_p/2$. What are the Fourier coefficients c_k for the signal $x_p(t)$?
(c) Using the results in parts (a) and (b), show that $c_k = (1/T_p)X_a(k/T_p)$.

4.4 Consider the following periodic signal:

$$x(n) = \{\ldots, 1, 0, 1, 2, \underset{\uparrow}{3}, 2, 1, 0, 1, \ldots\}$$

(a) Sketch the signal $x(n)$ and its magnitude and phase spectra.
(b) Using the results in part (a), verify Parseval's relation by computing the power in the time and frequency domains.

4.5 Consider the signal

$$x(n) = 2 + 2\cos\frac{\pi n}{4} + \cos\frac{\pi n}{2} + \frac{1}{2}\cos\frac{3\pi n}{4}$$

(a) Determine and sketch its power density spectrum
(b) Evaluate the power of the signal

4.6 Determine and sketch the magnitude and phase spectra of the following periodic signals.

(a) $x(n) = 4\sin\dfrac{\pi(n-2)}{3}$

(b) $x(n) = \cos\dfrac{2\pi}{3}n + \sin\dfrac{2\pi}{5}n$

(c) $x(n) = \cos\dfrac{2\pi}{3}n\sin\dfrac{2\pi}{5}n$

(d) $x(n) = \{\ldots, -2, -1, \underset{\uparrow}{0}, 1, 2, -2, -1, 0, 1, 2, \ldots\}$

(e) $x(n) = \{\ldots, -1, 2, \underset{\uparrow}{1}, 2, -1, 0, -1, 2, 1, 2, \ldots\}$

(f) $x(n) = \{\ldots, 0, 0, \underset{\uparrow}{1}, 1, 0, 0, 0, 1, 1, 0, 0, \ldots\}$

(g) $x(n) = 1, -\infty < n < \infty$

(h) $x(n) = (-1)^n, -\infty < n < \infty$

4.7 Determine the periodic signals $x(n)$, with fundamental period $N = 8$, if their Fourier coefficients are given by:

(a) $c_k = \cos\dfrac{k\pi}{4} + \sin\dfrac{3k\pi}{4}$

(b) $c_k = \begin{cases} \sin\dfrac{k\pi}{3}, & 0 \le k \le 6 \\ 0, & k = 7 \end{cases}$

(c) $\{c_k\} = \{\ldots, 0, \frac{1}{4}, \frac{1}{2}, 1, \underset{\uparrow}{2}, 1, \frac{1}{2}, \frac{1}{4}, 0 \ldots\}$

4.8 Two DT signals, $s_k(n)$ and $s_l(n)$, are said to be orthogonal over an interval $[N_1, N_2]$ if

$$\sum_{n=N_1}^{N_2} s_k(n)s_l^*(n) = \begin{cases} A_k, & k = l \\ 0, & k \ne l \end{cases}$$

If $A_k = 1$, the signals are called orthonormal.

(a) Prove the relation

$$\sum_{n=0}^{N-1} e^{j2\pi kn/N} = \begin{cases} N, & k = 0, \pm N, \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$$

(b) Illustrate the validity of the relation in part (a) by plotting for every value of $k = 1, 2, \ldots, 6$, the signals $s_k(n) = e^{j(2\pi/6)kn}$, $n = 0, 1, \ldots, 5$. *[Note:* For a given k, n the signal $s_k(n)$ can be represented as a vector in the complex plane.]

(c) Show that the harmonically related signals

$$s_k(n) = e^{j(2\pi/N)kn}$$

are orthogonal over any interval of length N.

4.9 Compute the Fourier transform of the following signals.

(a) $x(n) = u(n) - u(n-6)$

(b) $x(n) = 2^n u(-n)$

(c) $x(n) = (\frac{1}{4})^n u(n+4)$

(d) $x(n) = (a^n \sin\omega_0 n)u(n) \qquad |\alpha| < 1$

(e) $x(n) = |\alpha|^n \sin\omega_0 n \qquad |\alpha| < 1$

(f) $x(n) = \begin{cases} 2 - (\frac{1}{2})n, & |n| \le 4 \\ 0, & \text{elsewhere} \end{cases}$

(g) $x(n) = \{-2, -1, \underset{\uparrow}{0}, 1, 2\}$

(h) $x(n) = \begin{cases} A(2M + 1 - |n|), & |n| \le M \\ 0, & |n| > M \end{cases}$

Sketch the magnitude and phase spectra for parts (a), (f), and (g).

4.10 Determine the signals having the following Fourier transforms.

(a) $X(\omega) = \begin{cases} 0, & 0 \le |\omega| \le \omega_0 \\ 1, & \omega_0 < |\omega| \le \pi \end{cases}$

(b) $X(\omega) = \cos^2\omega$

(c) $X(\omega) = \begin{cases} 1, & \omega_0 - \delta\omega/2 \le |\omega| \le \omega_0 + \delta\omega/2 \\ 0, & \text{elsewhere} \end{cases}$

(d) The signal shown in Fig. P4.10.

4.11 Consider the signal

$$x(n) = \{1, 0, -1, \underset{\uparrow}{2}, 3\}$$

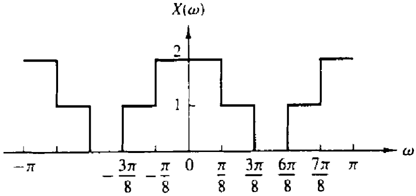


Figure P4.10

with Fourier transform $X(\omega) = X_R(\omega) + j(X_I(\omega))$. Determine and sketch the **signal** $y(n)$ with Fourier transform

$$Y(\omega) = X_I(\omega) + X_R(\omega)e^{j2\omega}$$

4.U Determine the signal $x(n)$ if its Fourier transform is as given in Fig. P4.12.

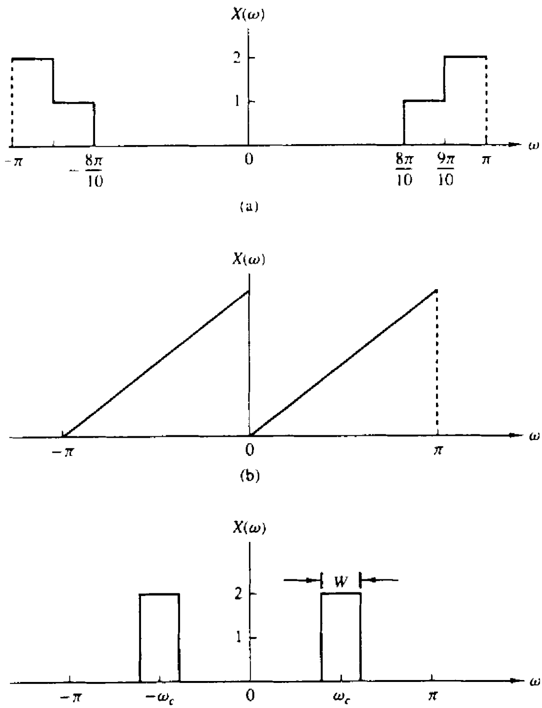


Figure P4.12

4.13 In Example 4.3.3, the Fourier transform of the signal

$$x(n) = \begin{cases} 1, & -M \le n \le M \\ 0, & \text{otherwise} \end{cases}$$

was shown to be

$$X(\omega) = 1 + 2\sum_{n=1}^{M} \cos \omega n$$

Show that the Fourier transform of

$$x_1(n) = \begin{cases} 1, & 0 \le n \le M \\ 0, & \text{otherwise} \end{cases}$$

and

$$x_2(n) = \begin{cases} 1, & -M \le n \le -1 \\ 0, & \text{otherwise} \end{cases}$$

are, respectively.

$$X_1(\omega) = \frac{1 - e^{-j\omega(M+1)}}{1 - e^{-j\omega}}$$

$$X_2(\omega) = \frac{e^{j\omega} - e^{j\omega(M+1)}}{1 - e^{j\omega}}$$

Thus prove that

$$\begin{aligned} X(\omega) &= X_1(\omega) + X_2(\omega) \\ &= \frac{\sin(M + \frac{1}{2})\omega}{\sin(\omega/2)} \end{aligned}$$

and therefore,

$$1 + 2\sum_{n=1}^{M} \cos \omega n = \frac{\sin(M + \frac{1}{2})\omega}{\sin(\omega/2)}$$

4.14 Consider the signal

$$x(n) = \{-1, 2, \underset{\uparrow}{-3}, 2, -1\}$$

with Fourier transform $X(\omega)$. Compute the following quantities, without explicitly computing $X(\omega)$:
(a) $X(0)$ **(b)** $\angle X(\omega)$ **(c)** $\int_{-\pi}^{\pi} X(\omega)\, d\omega$ **(d)** $X(\pi)$
(e) $\int_{-\pi}^{\pi} |X(\omega)|^2\, d\omega$

4.15 The center of gravity of a signal $x(n)$ is defined **as**

$$c = \frac{\sum_{n=-\infty}^{\infty} n x(n)}{\sum_{n=-\infty}^{\infty} x(n)}$$

and provides a measure of **the** "time delay" of the signal.

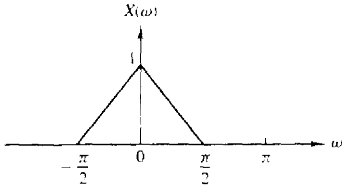


Figure P4.15

(a) Express c in terms of $X(\omega)$.
(b) Compute c lor the signal $x(n)$ whose Fourier transform is shown in Fig. P4.15

4.16 Consider the Fourier transform parr

$$a^n u(n) \; [redacted] \; \frac{1}{1 - ae^{-j\omega}} \qquad |a| < 1$$

Use the differentiation in frequency theorem and induction to show that

$$x(n) = \frac{(n+l-1)!}{n!(l-1)!} a^n u(n) \xleftrightarrow{F} X(\omega) = \frac{1}{(1 - ae^{-j\omega})^l}$$

4.17 Let $x(n)$ be an arbitrary signal, not necessarily real-valued, with Fourier transform $X(\omega)$. Express the Fourier transforms of the following signals in terms of $X(\omega)$.

(a) $x^*(n)$
(b) $x^*(-n)$
(c) $y(n) = x(n) - x(n-1)$
(d) $y(n) = \sum_{k=-\infty}^{n} x(k)$
(e) $y(n) = x(2n)$
(f) $y(n) = \begin{cases} x(n/2), & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$

4.18 Determine and sketch the Fourier transforms $X_1(\omega)$, $X_2(\omega)$, and $X_3(\omega)$ of the following signals.

(a) $x_1(n) = \{1, 1, 1, 1, 1\}$

(b) $x_2(n) = \{1, 0, 1, 0, \underset{\uparrow}{1}, 0, 1, 0, 1\}$

(c) $x_3(n) = \{1, 0, 0, 1, 0, 0, \underset{\uparrow}{1}, 0, 0, 1, 0, 0, 1\}$

(d) Is there any relation between $X_1(\omega)$, $X_2(\omega)$, and $X_3(\omega)$? What is its physical meaning?

(e) Show that if

$$x_k(n) = \begin{cases} x\left(\frac{n}{k}\right), & \text{if } n/k \text{ integer} \\ 0, & \text{otherwise} \end{cases}$$

then

$$X_k(\omega) = X(k\omega)$$

4.19 Let $x(n)$ be a signal with Fourier transform as shown in Fig. P4.19. Determine and sketch the Fourier transforms of the following signals.

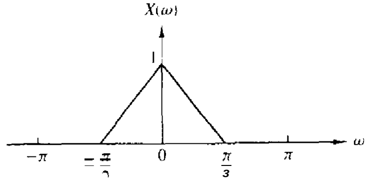


Figure P4.19

(a) $x_1(n) = x(n)\cos(\pi n/4)$ (b) $x_2(n) = x(n)\sin(\pi n/2)$
(c) $x_3(n) = x(n)\cos(\pi n/2)$ **(d)** $x_4(n) = x(n)\cos \pi n$

Note that these signal sequences are obtained by *amplitude modulation* of a carrier $\cos \omega_c n$ or $\sin \omega_c n$ by the sequence $x(n)$.

4.20 Consider an aperiodic signal $x(n)$ with Fourier transform $X(\omega)$. Show that the Fourier series coefficients C_k^y of the periodic signal

$$y(n) = \sum_{l=-\infty}^{\infty} x(n - lN)$$

are given by

$$C_k^y = \frac{1}{N} X\left(\frac{2\pi}{N}k\right) \qquad k = 0, 1, \ldots, N-1$$

4.21 Prove that

$$X_N(\omega) = \sum_{n=-N}^{N} \frac{\sin \omega_c n}{\pi n} e^{-j\omega n}$$

may be expressed as

$$X_N(\omega) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} \frac{\sin[(2N+1)(\omega - \theta/2)]}{\sin[(\omega - \theta)/2]} d\theta$$

4.22 A signal $x(n)$ has the following Fourier transform:

$$X(\omega) = \frac{1}{1 - ae^{-j\omega}}$$

Determine the Fourier transforms of the following signals:

(a) $x(2n+1)$ **(b)** $e^{\pi n/2} x(n+2)$
(b) $x(-2n)$ **(d)** $x(n)\cos(0.3\pi n)$
(c) $x(n) * x(n-1)$ **(f)** $x(n) * x(-n)$

4.23 From a discrete-time signal $x(n)$ with Fourier transform $X(\omega)$, shown in Fig. P4.23, determine and sketch the Fourier transform of the following signals:

(a) $y_1(n) = \begin{cases} x(n), & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$

(b) $y_2(n) = x(2n)$

(c) $y_3(n) = \begin{cases} x(n/2), & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$

Note that $y_1(n) = x(n)s(n)$, where $s(n) = \{\ldots 0, 1, 0, \underset{\uparrow}{1}, 0, 1, 0, 1, \ldots\}$

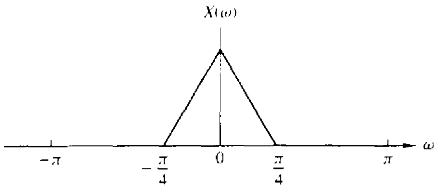


Figure P4.23

4.24 The following input–output pairs have been observed during the operation of various systems:

(a) $x(n) = (\frac{1}{2})^n \xrightarrow{T_1} y(n) = (\frac{1}{8})^n$

(b) $x(n) = (\frac{1}{2})^n u(n) \xrightarrow{T_2} y(n) = (\frac{1}{8})^n u(n)$

(c) $x(n) = e^{j\pi/5} \xrightarrow{T_3} y(n) = 3e^{j\pi/5}$

(d) $x(n) = e^{j\pi/5}u(n) \xrightarrow{T_4} y(n) = 3e^{j\pi/5}u(n)$

(e) $x(n) = x(n + N_1) \xrightarrow{T_5} y(n) = y(n + N_2) \qquad N_1 \neq N_2, \quad N_1, N_2$ prime

Determine their frequency response if each of the above systems is LTI.

4.25 (a) Determine and sketch tho Fourier transform $W_R(\omega)$ of the rectangular sequence

$$w_R(n) = \begin{cases} 1, & 0 \leq n \leq M \\ 0, & \text{otherwise} \end{cases}$$

(b) Consider the triangular sequence

$$w_T(n) = \begin{cases} n, & 0 \leq n \leq M/2 \\ M - n, & M/2 < 2 \leq M \\ 0, & \text{otherwise} \end{cases}$$

Determine and sketch the Fourier transform $W_T(\omega)$ of $w_T(n)$ by expressing it as the convolution of a rectangular scquenco with itself.

(c) Consider the sequence

$$w_c(n) = \frac{1}{2}\left(1 + \cos\frac{2\pi n}{M}\right) w_R(n)$$

Determine and sketch $W_c(\omega)$ by using $W_R(\omega)$.

4.26 Consider an LTI system wirh impulse response $h(n) = (\frac{1}{2})^n u(n)$.

(a) Determine and sketch the magnitude and phase response $|H(\omega)|$ and $\angle H(\omega)$, respectively.

(b) Determine and sketch tho magnitude and phase spectra for the input and output signals for the following inputs:

(1) $x(n) = \cos\dfrac{3\pi n}{10}, \; -\infty < n < \infty$

(2) $x(n) = \{\ldots, 1, 0, 0, \underset{\uparrow}{1}, 1, 1, 0, 1, 1, 1, 0, 1, \ldots\}$

4.27 Determine and skerch the magnitude and phase response of the following systems:

(a) $y(n) = \frac{1}{2}[x(n) + x(n-1)]$

(b) $y(n) = \frac{1}{2}[x(n) - x(n-1)]$

(c) $y(n) = \frac{1}{2}[x(n+1) - x(n-1)]$

(d) $y(n) = \frac{1}{2}[x(n+1) + x(n-1)]$

(e) $y(n) = \frac{1}{5}[x(n) + x(n-2)]$

(f) $y(n) = \frac{1}{2}[x(n) - x(n-2)]$

(g) $y(n) = \frac{1}{3}[x(n) + x(n-1) + x(n-2)]$

(h) $y(n) = x(n) - x(n-8)$

(i) $y(n) = 2x(n-1) - x(n-2)$

(j) $y(n) = \frac{1}{4}[x(n) + x(n-1) + x(n-2) + x(n-3)]$

(k) $y(n) = \frac{1}{8}[x(n) + 3x(n-1) + 3x(n-2) + x(n-3)]$

(l) $y(n) = x(n-4)$

(m) $y(n) = x(n+4)$

(n) $y(n) = \frac{1}{4}[x(n) - 2x(n-1) + x(n-2)]$

4.28 An FIR filter is described by the difference equation

$$y(n) = x(n) + x(n-10)$$

(a) Compute and sketch its magnitude and phase response.

(b) Determine its response to the inputs

(1) $x(n) = \cos\frac{\pi}{10}n + 3\sin\left(\frac{\pi}{3}n + \frac{\pi}{10}\right) \qquad -\infty < n < \infty$

(2) $x(n) = 10 + 5\cos\left(\frac{2\pi}{5}n + \frac{\pi}{2}\right) \qquad -\infty < n < \infty$

4.29 Determine the transient and steady-state responses of the FIR *filter* shown in Fig. *P4.29* to the input signal $x(n) = 10e^{j\pi n/2}u(n)$. Let $b = 2$ and $y(-1) = y(-2) = y(-3) = y(-4) = 0$.

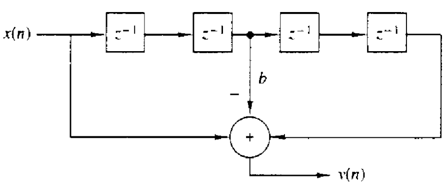


Figure P4.29

4.30 Consider the FIR filter

$$y(n) = x(n) + x(n-4)$$

(a) Compute and sketch its magnitude and phase response.

(b) Compute its response to the input

$$x(n) = \cos\frac{\pi}{2}n + \cos\frac{\pi}{4}n \qquad -\infty < n < \infty$$

(c) Explain the results obtained in part (b) in terms of the magnitude and phase responses obtained in part (a).

4.31 Determine the steady-state and transient responses of the system

$$y(n) = \frac{1}{2}[x(n) - x(n-2)]$$

to the input signal

$$x(n) = 5 + 3\cos\left(\frac{\pi}{2}n + 60^\circ\right) \qquad -\infty < n < \infty$$

4.32 From our discussions it is apparent thar an LTI sysrem cannot produce frequencies at its output that are different from those applied in its input. Thus, if a system creates "new" frequencies. it must be nonlinear and/or time varying. Determine the frequency content of the outputs of the following systems to the input signal

$$x(n) = A\cos\frac{\pi}{4}n$$

(a) $y(n) = x(2n)$
(b) $y(n) = x^2(n)$
(c) $y(n) = (\cos \pi n)x(n)$

4.33 Determine and sketch the magnitude and phase response of the systems shown in Fig. P4.33(a) through (c).

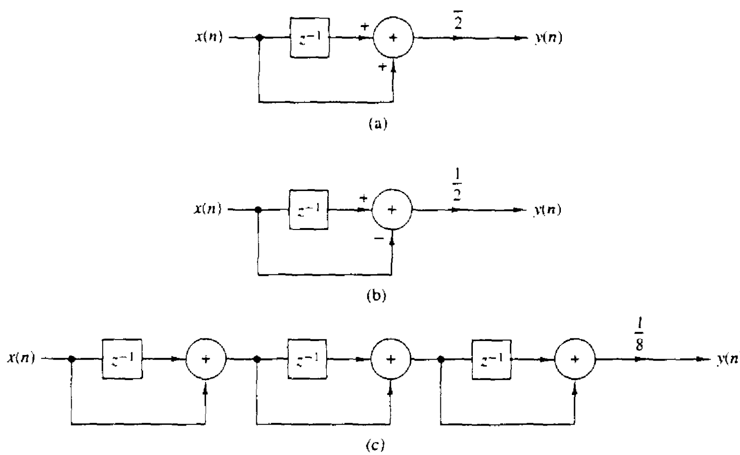


Figure P4.33

4.34 Determine the magnitude and phase response of the multipath channel

$$y(n) = x(n) + x(n - M)$$

At what frequencies does $H(\omega) = 0$?

4.35 Consider the filter

$$y(n) = 0.9y(n-1) + bx(n)$$

(a) Determine b so that $|H(0)| = 1$.
(b) Determine the frequency at which $|H(\omega)| = 1/\sqrt{2}$.

(c) Is this filter lowpass, bandpass. or highpass?
(d) Repeat parts (b) and (c) for the filter $y(n) = -0.9y(n-1) + 0.1x(n)$

4.36* ***Harmonic distortion in digital sinusoidal generators*** An ideal sinusoidal generator produces the signal

$$x(n) = \cos 2\pi f_0 n \qquad -\infty < n < \infty$$

which is periodic with fundamental period N if $f_0 = k_0/N$ and k_0, N are relatively prime numbers. The spectrum of such a "pure" sinusoid consist of two lines at $k = k_0$ and $k = N - k_0$ (we limit ourselves in the fundamental interval $0 \le k \le N - 1$). In practice, the approximations made in computing the samples of a sinusoid of relative frequency f_0 result in a certain amount of power falling into other frequencies. This spurious power results in distortion, which is referred to as ***harmonic distortion.*** Harmonic distortion is usually measured in terms of the *total harmonic* ***distortion*** (THD). which is defined as the ratio

$$\text{THD} = \frac{\text{spurious harmonic power}}{\text{total power}}$$

(a) Show that

$$\text{THD} = 1 - 2\frac{|c_{k_0}|^2}{P_x}$$

where

$$c_{k_0} = \frac{1}{N}\sum_{n=0}^{N-1} x(n)e^{-j(2\pi/N)k_0 n}$$

$$P_x = \frac{1}{N}\sum_{n=0}^{N-1} |x(n)|^2$$

(b) By using the Taylor approximation

$$\cos\phi = 1 - \frac{\phi^2}{2!} + \frac{\phi^4}{4!} - \frac{\phi^6}{6!} + \cdot$$

compute one period of $x(n)$ for $f_0 = 1/96$. 1132, 11256 by increasing the number of terms in the Taylor expansion from 2 to 8.

(c) Compute the THD and plot the power density spectrum for each sinusoid in part (b) as well as for the sinusoids obtained using the computer cosine function. Comment on the results.

4.37* ***Measurement of the total harmonic distortion in quantized sinusoids*** Let $x(n)$ be a periodic sinusoidal signal with frequency $f_0 = k/N$, that is.

$$x(n) = \sin 2\pi f_0 n$$

(a) Write a computer program that quantizes the signal $x(n)$ into b bits or equivalently into $L = 2^b$ levels by using rounding. The resulting signal is denoted by $x_q(n)$.
(b) For $f_0 = 1/50$ compute the THD of the quantized signals $x_q(n)$ obtained by using $b = 4$, 6, 8, and 16 bits.
(c) Repeat part **(b)** for $f_0 = 1/100$.
(d) Comment on the results obtained in parts (b) and (c).

4.38* Consider the discrete-time system

$$y(n) = ay(n-1) + (1-a)x(n) \qquad n \ge 0$$

where $a = 0.9$ and $y(-1) = 0$.

(a) Compute and sketch the output $y_i(n)$ of the system to the input signals

$$x_i(n) = \sin 2\pi f_i n \qquad 0 \le n \le 100$$

where $f_1 = \frac{1}{4}$, $f_2 = \frac{1}{5}$, $f_3 = \frac{1}{10}$, $f_4 = \frac{1}{20}$.

(b) Compute and sketch the magnitude and phase response of the svstem and use these results to explain the response of the svstem to the signals given in part (a).

4.39* Consider an LTI system with impulse response $h(n) = (\frac{1}{3})^{|n|}$

(a) Dcterminc and sketch the magnitude and phase response $|H(\omega)|$ and $\measuredangle H(\omega)$, respectively.

(b) Dcterminc and sketch the magnitude and phase spectra for the input and output signals for tho following inputs:

(1) $x(n) = \cos \dfrac{3\pi n}{8}, \; -\infty < n < \infty$

(2) $x(n) = \{\ldots, -1, 1, -1, \underset{\uparrow}{1}, -1, 1, -1, 1, -1, 1, -1, 1, \ldots\}$

4.40* *Time-domain sampling* Consider the continuous-time signal

$$x_a(t) = \begin{cases} e^{-j2\pi F_0 t}, & t \ge 0 \\ 0, & t < 0 \end{cases}$$

(a) Compute analytically the spectrum $X_a(F)$ of $x_a(t)$.

(b) Compute analytically the spectrum of the signal $x(n) = x_a(nT)$, T = 1/F,.

(c) Plot the magnitude spectrum $|X_a(F)|$ for $F_0 = 10$ *Hz*.

(d) Plot the magnitude spectrum $|X(F)|$ for $F_s = 10, 20,$ *40, and* 100 *Hz*.

(e) Explain the results obtained in part (d) in terms of the aliasing effect.

4.41 Consider the digital filter shown in Fig. P4.41.

(a) Determine the input–output relation and the impulse response $h(n)$.

(b) Determine and akctch the magnitude $|H(\omega)|$ and the phase response $\measuredangle H(\omega)$ of the filter and find which frequencies are completely blocked by the filter.

(c) When $\omega_0 = \pi/2$, determine the output $y(n)$ to the input

$$x(n) = 3\cos\left(\frac{\pi}{3}n + 30\right) \qquad -\infty < n < \infty$$

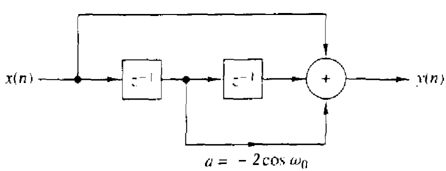


Figure P4.41

4.42 Consider the FIR filter

$$y(n) = x(n) - x(n-4)$$

(a) Compute and sketch its magnitude and phase response

(h) Compute its response to the input

$$x(n) = \cos\frac{\pi}{2}n + \cos\frac{\pi}{4}n \qquad -\infty < n < \infty$$

(c) Explain the results obtained in part (b) in terms of the answer given in part (a).

4.43 Determine the steadv-state response of the system

$$y(n) = \tfrac{1}{2}[x(n) - x(n-2)]$$

to the input signal

$$x(n) = 5 + 3\cos\left(\frac{\pi}{2}n + 60^\circ\right) + 4\sin(\pi n + 45^\circ) \qquad -\infty < n < \infty$$

4.44 Recall from Problem 4.32 that an LTI system cannot produce frequencies at its output that are different from those applied in its input. Thus if a system creates **"new"** frequencies, it must be nonlinear and/or time varying. Indicate whether the following systems are nonlinear and/or time varying and determine the output spectra when the input spectrum is

$$X(\omega) = \begin{cases} 1, & |\omega| \le \pi/4 \\ 0, & \pi/4 \le |\omega| \le \pi \end{cases}$$

(a) $y(n) = x(2n)$
(b) $y(n) = x^2(n)$
(c) $y(n) = (\cos \pi n)x(n)$

4.45 Consider an LTI system with impulse response

$$h(n) = \left[\left(\frac{1}{4}\right)^n \cos\left(\frac{\pi}{4}n\right)\right] u(n)$$

(a) Determine its system function $H(z)$.
(b) Is it possible to implement this system using a finite number of adders. multipliers, and unit delays'.' If yes. how'?
(c) Provide a rough sketch of $|H(\omega)|$ using the pole–zero plot.
(d) Determine the response of the system to the input

$$x(n) = (\tfrac{1}{4})^n u(n)$$

4.46 An FIR filter is described by the difference equation

$$y(n) = x(n) - x(n-10)$$

(a) Compute and sketch its magnitude and phase response.
(b) Determine its response to the inputs

(1) $x(n) = \cos\frac{\pi}{10}n + 3\sin\left(\frac{\pi}{3}n + \frac{\pi}{10}\right) \qquad -\infty < n < \infty$

(2) $x(n) = 5 + 6\cos\left(\frac{2\pi}{5}n + \frac{\pi}{2}\right) \qquad -\infty < n < \infty$

4.47 The frequency response of an ideal bandpass filter is given by

$$H(\omega) = \begin{cases} 0, & |\omega| \le \frac{\pi}{8} \\ 1, & \frac{\pi}{8} < |\omega| < \frac{3\pi}{8} \\ 0, & \frac{3\pi}{8} \le |\omega| \le \pi \end{cases}$$

(a) Determine its impulse response
(b) Show that this impulse response can be expressed as the product of $\cos(n\pi/4)$ and the impulse response of a lowpass filter.

4.48 Consider the system described by the difference equation

$$y(n) = \tfrac{1}{2}y(n-1) + x(n) + \tfrac{1}{2}x(n-1)$$

(a) Determine its impulse response.
(b) Determine its frequency response:
 (1) From the impulse response
 (2) From the difference equation
(c) Determine its response to the input

$$x(n) = \cos\left(\frac{\pi}{2}n + \frac{\pi}{4}\right) \qquad -\infty < n < \infty$$

4.49 Sketch roughly the magnitude $|X(\omega)|$ of the Fourier transforms corresponding to the pole-zero patterns given in Fig. P4.49.

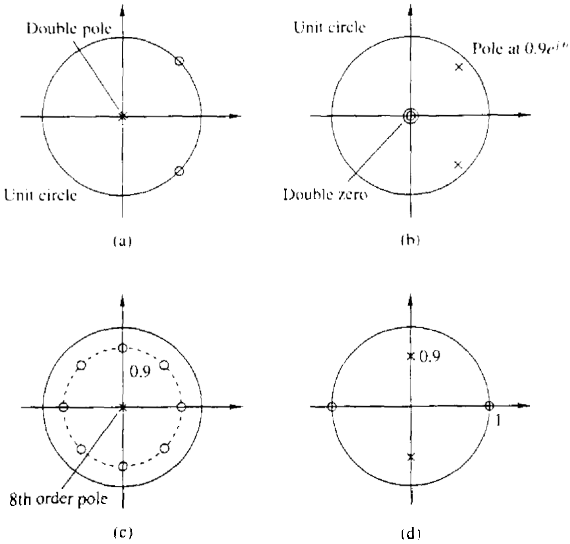


Figure P4.49

4.50 Design an FIR filter that completely blocks the frequency $\omega_0 = \pi/4$ and then compute its output if the input is

$$x(n) = \left(\sin \frac{\pi}{4}n\right) u(n)$$

for $n = 0, 1, 2, 3, 4$. Does the filter fulfill your expectations? Explain

4.51 A digital filter is characterized by the following properties:

(1) It is highpass and has one pole and one zero.
(2) The pole is at a distance $r = 0.9$ from the origin of the z-plane.
(3) Constant signals do not pass through the system.

(a) Plot the pole–zero pattern of the filter and determine its system function $H(z)$.
(b) Compute the magnitude response $|H(\omega)|$ and the phase response $\measuredangle H(\omega)$ of the filter.
(c) Normalize the frequency response $H(\omega)$ so that $|H(\pi)| = 1$.
(d) Determine the input–output relation (difference equation) of the filter in the time domain.
(e) Compute the output of the system if the input is

$$x(n) = 2\cos\left(\frac{\pi}{6}n + 45^\circ\right) \qquad -\infty < n < \infty$$

(You can use either algebraic or geometrical arguments.)

4.52 A causal first-order digital filter is described by the system function

$$H(z) = b_0 \frac{1 + bz^{-1}}{1 + az^{-1}}$$

(a) Sketch the direct form I and direct form II realizations of this filter and find the corresponding difference equations.
(b) For $a = 0.5$ and $b = -0.6$. sketch the pole–zero pattern. Is the system stable? Why?
(c) For $a = -0.5$ and $b = 0.5$, determine b_0, so that the maximum value of $|H(\omega)|$ is equal to 1.
(d) Sketch the magnitude response $|H(\omega)|$ and the phase response $\measuredangle H(\omega)$ of the filter obtained in part (c).
(e) In a specific application it is known that $a = 0.8$. Does the resulting filter amplify high frequencies or low frequencies in the input? Choose the value of b so as to improve the characteristics of this filter (i.e., make it a better lowpass or a better highpass filter).

453 Derive the expression for the resonant frequency of a two-pole filter with poles at $p_1 = re^{j\theta}$ and $p_2 = p_1^*$, given by (4.5.25).

4.54 Determine and sketch the magnitude and phase responses of the Hanning filter characterized by the (moving average) difference equation

$$y(n) = \tfrac{1}{4}x(n) + \tfrac{1}{2}x(n-1) + \tfrac{1}{4}x(n-2)$$

4.55 A causal LTI system excited by the input

$$x(n) = (\tfrac{1}{4})^n u(n) + u(-n-1)$$

produces an output $y(n)$ with z-transform

$$Y(z) = \frac{-\frac{3}{4}z^{-1}}{(1 - \frac{1}{4}z^{-1})(1 + z^{-1})}$$

(a) Determine the system function $H(z)$ and its ROC.
(b) Determine the output $y(n)$ of the system.
(Hint: Pole cancellation increases the original ROC.)

4.56 Determine the coefficients of a linear-phase FIR filter

$$y(n) = b_0x(n) + b_1x(n-1) + b_2x(n-2)$$

such that:

(a) It rejects completely a frequency component at $\omega_0 = 2\pi/3$.

(b) Its frequency response is normalized so that $H(0) = 1$.

(c) Compute and sketch the magnitude and phase response of the filter to check if it satisfies the requirements.

4.57 Determine the frequency response $H(\omega)$ of the following moving average filters.

(a) $y(n) = \dfrac{1}{2M+1}\displaystyle\sum_{k=-M}^{M} x(n-k)$

(b) $y(n) = \dfrac{1}{4M}x(n+M) + \dfrac{1}{2M}\displaystyle\sum_{k=-M+1}^{M-1} x(n-k) + \dfrac{1}{4M}x(n-M)$

Which filter provides better smoothing? Why?

4.58 The convolution $x(t)$ of two continuous-time signals $x_1(t)$ and $x_2(t)$, from which at least one is nonperiodic. is defined by

$$x(t) \triangleq x_1(t) * x_2(t) \triangleq \int_{-\infty}^{\infty} x_1(\lambda)x_2(t-\lambda)d\lambda$$

(a) Show that $X(F) = X_1(F)X_2(F)$, where $X_1(F)$ and $X_2(F)$ are the spectra of $x_1(t)$ and $x_2(t)$, respectively.

(b) Compute $x(t)$ if $x_1(t) = x_2(t) = \begin{cases} 1, & |t| < \tau/2 \\ 0, & \text{elsewhere} \end{cases}$

(c) Determine the spectrum of $x(t)$ using the results in part (a).

4.59 Compute the magnitude and phase response of a filter with system function

$$H(z) = 1 + z^{-1} + z^{-2} + \cdots + z^{-8}$$

If the sampling frequency is $F_s = 1$ **kHz**. determine the frequencies of the analog sinusoids that cannot pass through the filter.

4.60 A second-order system has a double **pole** at $p_{1,2} = 0.5$ and two zeros at

$$z_{1,2} = e^{\pm j3\pi/4}$$

Using geometric arguments, choose the gain G of the filter so that $|H(0)| = 1$.

4.61 In this problem we consider the effect of a single zero on the frequency response of a system. Let $z = re^{j\theta}$ be a zero inside the unit circle ($r < 1$). Then

$$\begin{aligned} H_z(\omega) &= 1 - re^{j\theta}e^{-j\omega} \\ &= 1 - r\cos(\omega-\theta) + jr\sin(\omega-\theta) \end{aligned}$$

(a) Show that the magnitude response is

$$|H_z(\omega)| = [1 - 2r\cos(\omega-\theta) + r^2]^{1/2}$$

or, equivalently.

$$20\log_{10}|H_z(\omega)| = 10\log_{10}[1 - 2r\cos(\omega-\theta) + r^2]$$

(b) Show that the phase response is given as

$$\Theta_z(\omega) = \tan^{-1}\frac{r\sin(\omega-\theta)}{1-r\cos(\omega-\theta)}$$

(c) Show that the group delay is given as

$$\tau_g^z(\omega) = \frac{r^2 - r\cos(\omega-\theta)}{1+r^2-2r\cos(\omega-\theta)}$$

(d) Plot the magnitude $|H(\omega)|_{dB}$, the phase $\Theta(\omega)$ and the group delay $\tau_g(\omega)$ for $r = 0.7$ and $\theta = 0, \pi/2$, and π.

4.62 In this problem we consider the effect of a single pole on the frequency response of a system. Hence, we let

$$H_p(\omega) = \frac{1}{1 - re^{j\theta}e^{-j\omega}} \qquad r < 1$$

Show that

$$\begin{aligned} |H_p(\omega)|_{dB} &= -|H_z(\omega)|_{dB} \\ \measuredangle H_p(\omega) &= -\measuredangle H_z(\omega) \\ \tau_g^p(\omega) &= -\tau_g^z(\omega) \end{aligned}$$

where $H_z(\omega)$ and $\tau_g^z(\omega)$ are defined in Problem 4.61.

4.63 In this problem we consider the effect of complex-conjugate pair of poles and zeros on the frequency response of a system. Let

$$H_z(\omega) = (1 - re^{j\theta}e^{-j\omega})(1 - re^{-j\theta}e^{-j\omega})$$

(a) Show that the magnitude response in decibels is

$$\begin{aligned} |H_z(\omega)|_{dB} = {} & 10\log_{10}[1 + r^2 - 2r\cos(\omega-\theta)] \\ & + 10\log_{10}[1 + r^2 - 2r\cos(\omega+\theta)] \end{aligned}$$

(b) Show that the phase response is given as

$$\Theta_z(\omega) = \tan^{-1}\frac{r\sin(\omega-\theta)}{1-r\cos(\omega-\theta)} + \tan^{-1}\frac{r\sin(\omega+\theta)}{1-r\cos(\omega+\theta)}$$

(c) Show that the group delay is given as

$$\tau_g^z(\omega) = \frac{r^2 - r\cos(\omega-\theta)}{1+r^2-2r\cos(\omega-\theta)} + \frac{r^2 - r\cos(\omega+\theta)}{1+r^2-2r\cos(\omega+\theta)}$$

(d) If $H_p(\omega) = 1/H_z(\omega)$, show that

$$\begin{aligned} |H_p(\omega)|_{dB} &= -|H_z(\omega)|_{dB} \\ \Theta_p(\omega) &= -\Theta_z(\omega) \\ \tau_g^p(\omega) &= -\tau_g^z(\omega) \end{aligned}$$

(e) Plot $|H_p(\omega)|$, $\Theta_p(\omega)$ and $\tau_g^p(\omega)$ for $r = 0.9$, and $\theta = 0, \pi/2$.

4.64 Determine the ?-dB bandwidth of the filters $(0 < a < 1)$

$$H_1(z) = \frac{1-a}{1-az^{-1}}$$

$$H_2(z) = \frac{1-a}{2}\frac{1+z^{-1}}{1-az^{-1}}$$

Which is a better lowpass filter?

4.65 Design a digital oscillator with adjustable phase. that is. a digital filter which produces the signal

$$y(n) = \cos(\omega_0 n + \theta)u(n)$$

4.66 This problem provides another derivation of the structure for the coupled-form oscillator by considering the system

$$y(n) = ay(n-1) + x(n)$$

for $a = e^{j\omega_0}$.

Let $x(n)$ be real. Then $y(n)$ is complex. Thus

$$y(n) = y_R(n) + jy_I(n)$$

(a) Determine the equations describing a system with one input $x(n)$ and the two outputs $y_R(n)$ and $y_I(n)$.

(b) Determine a block diagram realization

(c) Show that if $x(n) = \delta(n)$. then

$$y_R(n) = (\cos\omega_0 n)u(n)$$

$$y_I(n) = (\sin\omega_0 n)u(n)$$

(d) Compute $y_R(n)$. $y_I(n)$. $n = 0, 1, \ldots, Y$ for $\omega_0 = \pi/6$. Compare these with the true values **of** the sine and cosine.

4.67 Consider a filter with system function

$$H(z) = b_0\frac{(1-e^{j\omega_0}z^{-1})(1-e^{-j\omega_0}z^{-1})}{(1-re^{j\omega_0}z^{-1})(1-re^{-j\omega_0}z^{-1})}$$

(a) Sketch the pole–zcro pattern.

(b) Using geometric arguments. show that for $r \simeq 1$. the system is a notch filter and provide a rough sketch of its magnitude response if $\omega_0 = 60$.

(c) For $\omega_0 = 60$. choose b_0 so that the maximum value of $|H(\omega)|$ is 1.

(d) Draw a direct form II realization **of** the system

(e) Dctcrmine the approximate 3-dB bandwidth of the system,

4.68* Design an FIR digital filter that will reject a very strong 60-Hz sinusoidal interference contaminating a 200-Hz useful sinusoidal signal. Determine the gain of the filter so that the useful signal does not change amplitude. The filter works at a sampling frequency $F_s = 500$ samples/s. Compute the output of the filter if the input *is* a 60-Hz sinusoid or a 200-Hz sinusoid with unit amplitude. How does the performance **of** the filter compare with your requirements?

4.69 Determine the gain b_0 for the digital resonator described by (4.5.28) so that $|H(\omega_0)| = 1$.

4.70 Demonstrate that the difference equation given in (4.5.52) can be ohtained by applying the trigonometric identity

$$\cos\alpha + \cos\beta = 2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$$

where $\alpha = (n+1)\omega_0$, $\beta = (n-1)\omega_0$, and $y(n) = \cos\omega_0 n$. Thus show that the sinusoidal signal $y(n) = A\cos\omega_0 n$ can be generared from (4.5.52) hy use of the initial conditions $y(-1) = A\cos\omega_0$ and $y(-2) = A\cos 2\omega_0$.

4.71 Use the trigonometric identity in (4.5.53) with $\alpha = n\omega_0$ and $\beta = (n-2)\omega_0$ to derive the difference equation for generating the sinusoidal signal $y(n) = A\sin n\omega_0$. Determine the corresponding initial conditions.

4.72 Using the z-transform pairs 8 and 9 in Tahle 3.3, determine the difference equations for the digital oscillators that have impulse responses $h(n) = A\cos n\omega_0 u(n)$ and $h(n) = A\sin n\omega_0 u(n)$, respectively.

4.73 Determine the structure for the coupled-form oscillator by combining the structure for the digital oscillators obtained in Problem 4.72.

4.74 Convert the highpass filter with system function

$$H(z) = \frac{1-z^{-1}}{1-az^{-1}} \qquad a < 1$$

into a notch filter that rejects the frequency $\omega_0 = \pi/4$ and its harmonics.

(a) Determine the difference equation.

(b) Sketch the pole-zero pattern.

(c) Sketch the magnitude response for both filters.

4.75 Choose L and M for a lunar filter that must have narrow passbands at $(k \pm \Delta F)$ cycles/day, where $k = 1, 2, 3, \ldots$ and $\Delta F = 0.067726$.

4.76 **(a)** Show that the systems corresponding to the pole–zero patterns of Fig. 4.58 are all-pass.

(b) What is the number of delays and multipliers required for the efficient implementation of a second-order all-pass system'!

4.77 **A** digital notch filter is required to remove an undesirable 60-Hz hum associated with a power supply in an ECG recording application. The sampling frequency used is $F_s = 500$ samples/s. **(a)** Design a second-order FIR notch filter and **(b)** a second-order pole–zero notch filter for this purpose. In hoth cases choose the gain b_0 so that $|H(\omega)| = 1$ for $\omega = 0$.

4.78 Determine the coefficients $\{h(n)\}$ of a highpass linear phase FIR filter of length $M = 4$ which has an antisymmetric unit sample response $h(n) = -h(M-1-n)$ and a frequency response that satisfies the condition

$$\left|H\left(\frac{\pi}{4}\right)\right| = \frac{1}{2} \qquad \left|H\left(\frac{3\pi}{4}\right)\right| = 1$$

4.79 In an attempt to design a four-pole bandpass digital filter with desired magnitude response

$$|H_d(\omega)| = \begin{cases} 1, & \dfrac{\pi}{6} \le \omega \le \dfrac{\pi}{2} \\ 0, & \text{elsewhere} \end{cases}$$

we select the four poles at

$$p_{1,2} = 0.8e^{\pm j2\pi/9}$$
$$p_{3,4} = 0.8e^{\pm j4\pi/9}$$

and four zeros at

$$z_1 = 1 \qquad = -1 \qquad z_{3,4} = e^{\pm j3\pi/4}$$

(a) Determine the value of the gain so that

$$\left| H\left(\frac{5\pi}{12}\right) \right| = 1$$

(b) Determine the system function $H(z)$.

(c) Determine the magnitude of the frequency response $H(\omega)$ for $0 \le \omega \le \pi$ and compare it with the desired response $|H_d(\omega)|$.

4.80 A discrete-time system with input $x(n)$ and output $y(n)$ is described in the frequency domain by the relation

$$Y(\omega) = e^{-j2\pi\omega}X(\omega) + \frac{dX(\omega)}{d\omega}$$

(a) Compute the response of the system to the input $x(n) = \delta(n)$.

(b) Check if the system is LTI and stable.

4.81 Consider an ideal lowpass filter with impulse response $h(n)$ and frequency response

$$H(\omega) = \begin{cases} 1, & |\omega| \le \omega_c \\ 0, & \omega_c < |\omega| < \pi \end{cases}$$

What is the frequency response of the filter defined by

$$g(n) = \begin{cases} h\left(\frac{n}{2}\right), & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$$

4.82 Consider the system shown in Fig. P4.82. Determine its impulse response and its frequency response if the system $H(\omega)$ is:

(a) Lowpass with cutoff frequency ω_c.

(b) Highpass with cutoff frequency ω_c.

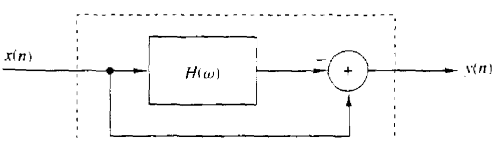


Figure P4.82

4.83 Frequency inverters have been used for many years for speech scrambling. Indeed, a voice signal $x(n)$ becomes unintelligible if we invert its spectrum as shown in Fig. P4.83.

(a) Determine how frequency inversion can be performed in the time domain.

(b) Design an unscrambler. *(Hint:* The required operations are very simple and can easily be done in real time.)

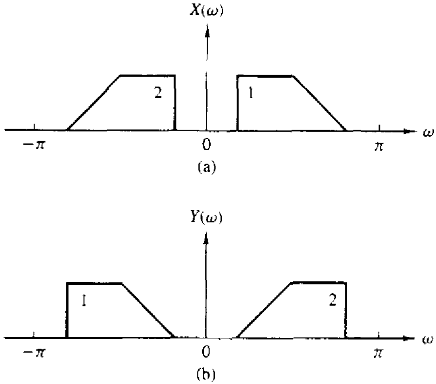


Figure P4.83 (a) Original spectrum; (b) frequency-inverted spectrum.

4.84 A lowpass filter is described by the difference equation

$$y(n) = 0.9y(n-1) + 0.1x(n)$$

(a) By performing a frequency translation of $\pi/2$, transform the filter into a bandpass filter.
(b) What is the impulse response of the bandpass filter?
(c) What is the major problem with the frequency translation method for transforming a prototype lowpass filter into a bandpass filter?

4.85 Consider a system with a real-valued impulse response $h(n)$ and frequency response

$$H(\omega) = |H(\omega)|e^{j\theta(\omega)}$$

The quantity

$$D = \sum_{n=-\infty}^{\infty} n^2 h^2(n)$$

provides a measure of the "effective duration" of $h(n)$.
(a) Express D in terms of $H(\omega)$.
(b) Show that D is minimized for $\theta(\omega) = 0$.

4.86 Consider the lowpass filter

$$y(n) = ay(n-1) + bx(n) \qquad 0 < a < 1$$

(a) Determine b so that $|H(0)| = 1$.
(b) Determine the 3-dB bandwidth ω_3 for the normalized filter in part (a).
(c) How does the choice of the parameter a affect ω_3?
(d) Repeat parts (a) through (c) for the highpass filter obtained by choosing $-1 < a < 0$.

4.87 Sketch the magnitude and phase response of the multipath channel

$$y(n) = x(n) + \alpha x(n-M) \qquad \alpha > 0$$

for $\alpha \ll 1$.

4.88 Determine the system functions and the pole-zero locations for the systems shown in Fig. P4.88(a) through (c). and indicate whether or not the systems are stable.

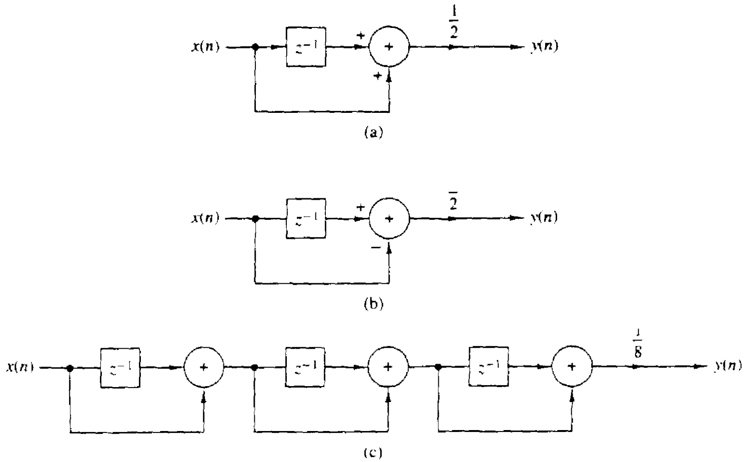


Figure P4.88

4.89 Determine and sketch the impulse response and the magnitude and phase responses of the FIR filter shown in Fig, P4.89 for $b = 1$ and $b = -1$.

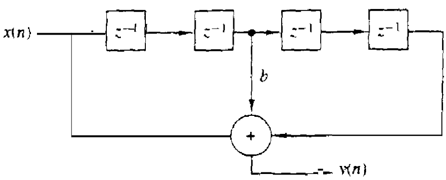


Figure P4.89

4.90 Consider the system

$$y(n) = x(n) - 0.95x(n-6)$$

(a) Sketch its pole-zero pattern.
(b) Sketch its magnitude response using the pole–zero plot.
(c) Determine the system function of its causal inverse system.
(d) Sketch the magnitude response of the inverse system using the pole–zero plot.

4.91 Determine the impulse response and the difference equation for all possible systems specified by the system functions

(a) $H(z) = \dfrac{z^{-1}}{1 - z^{-1} - z^{-2}}$

(b) $H(z) = \dfrac{1}{1 - e^{-4a}z^{-4}} \qquad 0 < a < 1$

4.92 Determine the impulse response of a causal LTI system which produces the response

$$y(n) = \{\underset{\uparrow}{1}, -1, 3, -1, 6\}$$

when excited by the input signal

$$x(n) = \{\underset{\uparrow}{1}, 1, 2\}$$

4.93 The system

$$y(n) = \tfrac{1}{2}y(n-1) + x(n)$$

is excited with the input

$$x(n) = (\tfrac{1}{4})^n u(n)$$

Determine the sequences $r_{xx}(l)$, $r_{hh}(l)$, $r_{xy}(l)$, and $r_{yy}(l)$.

4.94 Determine if the following FIR systems are minimum phase

(a) $h(n) = \{\underset{\uparrow}{10}, 9, -7, -8, 0, 5, 3\}$

(b) $h(n) = \{\underset{\uparrow}{5}, 4, -3, -4, 0, 2, 1\}$

4.95 Can you determine the coefficients of the all-pole system

$$H(z) = \frac{1}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

if you know its order N and the values $h(0), h(1), \ldots, h(L-1)$ of its impulse response? How? What happens if you do not know N?

4.96 Consider a system with impulse response

$$h(n) = b_0\delta(n) + b_1\delta(n-D) + b_2\delta(n-2D)$$

(a) Explain why the system generates echoes spaced D samples apart.

(b) Determine the magnitude and phase response of the system.

(c) Show that for $|b_0 + b_2| << |b_1|$, the locations of maxima and minima of $|H(\omega)|^2$ are at

$$\omega = \pm\frac{k}{D}\pi \qquad k = 0, 1, 2, \ldots$$

(d) Plot $|H(\omega)|$ and $\measuredangle H(\omega)$ for $b_0 = 0.1$, $b_1 = 1$, and $b_2 = 0.05$ and discuss the results.

4.97 Consider the pole-zero system

$$H(z) = \frac{B(z)}{A(z)} = \frac{1 + bz^{-1}}{1 + az^{-1}} = \sum_{n=0}^{\infty} h(n)z^{-n}$$

(a) Determine $h(0)$, $h(1)$, $h(2)$, and $h(3)$ in terms of a and b.

(b) Let $r_{hh}(l)$ be the autocorrelation sequence of $h(n)$. Determine $r_{hh}(0)$, $r_{hh}(1)$, $r_{hh}(2)$, and $r_{hh}(3)$ in terms of a and b.

4.98 Let $x(n)$ be a real-valued minimum-phase sequence. Modify $x(n)$ to obtain another real-valued minimum-phase sequence $y(n)$ such that $y(0) = x(0)$ and $y(n) = |x(n)|$.

4.99 The frequency response of a stable LTI system is known to be real and even. Is the inverse system stable?

4.100 Let $h(n)$ he a real filter with nonzero linear or nonlinear phase response. Show that the following operations are equivalent to filtering the signal $x(n)$ with a zero-phase filter.

(a) $g(n) = h(n) * x(n)$
$f(n) = h(n) * g(-n)$
$y(n) = f(-n)$

(b) $g(n) = h(n) * x(n)$
$f(n) = h(n) * x(-n)$
$y(n) = g(n) + f(-n)$

(Hint: Determine the frequency response of the composite system $y(n) = H[x(n)]$.)

4.101 Check the validity of the following statements:

(a) The convolution of two minimum-phase sequences is always minimum-phase sequence.

(b) The sum of two minimum-phase sequences is always minimum phase.

4.102 Determine the minimum-phase system whose squared magnitude response is given by:

(a) $|H(\omega)|^2 = \dfrac{\frac{5}{4} - \cos\omega}{\frac{10}{9} - \frac{2}{3}\cos\omega}$

(b) $|H(\omega)|^2 = \dfrac{2(1 - a^2)}{(1 + a^2) - 2a\cos\omega} \qquad |a| < 1$

4.103 Consider an FIR system with the following system function:

$$H(z) = (1 - 0.8e^{j\pi/2}z^{-1})(1 - 0.8e^{-j\pi/2}z^{-1})(1 - 1.5e^{j\pi/4}z^{-1})(1 - 1.5e^{-j\pi/4}z^{-1})$$

(a) Determine all systems that have the same magnitude response. Which is the minimum-phase system?

(b) Determine the impulse response of all systems in part (a).

(c) Plot the partial energy

$$E(n) = \sum_{k=0}^{n} h^2(n)$$

for every system and use it to identify the minimum- and maximum-phase systems.

4.104 The causal system

$$H(z) = \frac{1}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

is known to he unstable.

We modify this system by changing its impulse response $h(n)$ to

$$h'(n) = \lambda^n h(n)u(n)$$

(a) Show that by properly choosing λ we can obtain a new stable system.
(b) What is the difference equation describing the new system?

4.105* Given a signal $x(n)$, we can create echoes and reverberations by delaying and scaling the signal as follows

$$y(n) = \sum_{k=0}^{\infty} g_k x(n - kD)$$

where D is positive integer and $g_k > g_{k-1} > 0$.

(a) Explain why the comb filter

$$H(z) = \frac{1}{1 - az^{-D}}$$

can he used as a reverberator (i.e.. as a device to produce artificial reverberations). ***(Hint:*** Determine and sketch its impulse response.)

(b) The all-pass comb filter

$$H(z) = \frac{z^{-D} - a}{1 - az^{-D}}$$

is used in practice to build digital reverberators by cascading three to five such fillers and properly choosing the parameters a and D. Compute and plot the impulse response of two such reverberators each obtained by cascading three sections with the following parameters.

UNIT 1

Section	D	a
1	50	0.7
2	40	0.665
3	32	0.63175

UNIT 2

Section	D	a
1	50	0.7
2	17	0.77
3	6	0.847

(c) The difference between echo and reverberation is that with pure echo there are clear repetitions of the signal. but with reverberations, there are not. How is this reflected in the shape of the impulse response of the reverberator? Which unit in part (b) is a better reverberator?

(d) If the delays D_1, D_2, D_3 in a certain unit are prime numbers, the impulse response of the unit is more "dense." Explain why.

(e) Plot the phase response of units 1 and 2 and comment on them.

(f) Plot $h(n)$ for D_1, D_2, and D_3 being nonprime. What do you notice?

More details about this application can be found in a paper by J. A. Moorer, "Signal Processing Aspects of Computer Music: **A** Survey," *Proc*, IEEE, vol. 65, No. 8, Aug. 1977. pp. 1108–1137.

4.106* By trial-and-error design a third-order lowpass filter with cutoff frequency at $\omega_c = \pi/9$ radianstsample interval. Start your search with

(a) $z_1 = z_2 = z_3 = 0$, $p_1 = r$, $p_{2,3} = re^{\pm j\omega_c}$, $r = 0.8$

(b) $r = 0.9$, $z_1 = z_2 = z_3 = -1$

4.107* A speech signal with bandwidth B = 10 kHz is sampled at F_s = 20 kHz. Suppose that the signal is corrupted by four sinusoids with frequencies

$$F_1 = 10,000 \text{ Hz}, \qquad F_3 = 7778 \text{ Hz}$$

$$F_2 = 8889 \text{ Hz}, \qquad F_4 = 6667 \text{ Hz}$$

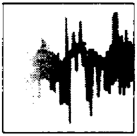
(a) Design a FIR filter that eliminates these frequency components.
(b) Choose the gain of the filter so that $|H(0)| = 1$ and then plot the log magnitude response and the phase response of the filter.
(c) Does the filter fulfill your objectives? Do you recommend the use of this filter in a practical application?

4.108* Compute and sketch the frequency response of a digital resonator with $\omega = \pi/6$ and r = 0.6. 0.9, 0.99. In each case. compute the bandwidth and the resonance frequency from the graph, and check if they are in agreement with the theoretical results.

4.109* The system function of a communication channel is given by

$$H(z) = (1 - 0.9e^{j0.4\pi}z^{-1})(1 - 0.9e^{-j0.4\pi}z^{-1})(1 - 1.5e^{j0.6\pi}z^{-1})(1 - 1.5e^{-j0.6\pi}z^{-1})$$

Determine the system function $H_c(z)$ of a causal and stable compensating system so that the cascade interconnection of the two systems has a flat magnitude response. Sketch rhc pole–zero plots and the magnitude and phase responses of all systems involved into the analysis process. [*Hint:* Use the decomposition $H(z) = H_{ap}(z)H_{min}(z)$.]



5

The Discrete Fourier Transform: Its Properties and Applications

Frequency analysis of discrete-time signals is usually and most conveniently performed on a digital signal processor. which may be a general-purpose digital corn-puter or specially designed digital hardware. To perform frequency analysis on a discrete-time signal $\{x(n)\}$. we convert tho time-domain scqucncc to an equivalent frequency-domain representation. Wc know that such a representation is given by the Fourier transform $X(\omega)$ of the scqucncc $\{x(n)\}$. However. $X(\omega)$ is a continuous function of frequency and therefore. it is not a computationally convenient representation of the sequence $\{x(n)\}$.

In this section we consider the representation of a sequence $\{x(n)\}$ by samples of its spectrum $X(\omega)$. Such a frequency-domain representation leads to the discrete Fourier transform (DFT). which is a powerful computational tool for performing frequency analvsis of discrete-time signals.

5.1 FREQUENCY DOMAIN SAMPLING: THE DISCRETE FOURIER TRANSFORM

Before we introduce the DFT, we consider the sampling of the Fourier transform of an aperiodic discrete-time sequence. Thus. we establish the relationship between the sampled Fourier transform and the DFT.

5.1.1 Frequency-Domain Sampling and Reconstruction of Discrete-Time Signals

We recall that aperiodic finite-energy signals have continuous spectra. Let us consider such an aperiodic discrete-time signal $x(n)$ with Fourier transform

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \tag{5.1.1}$$

Suppose that we sample $X(\omega)$ periodically in frequency at a spacing of $\delta\omega$ radians between successive samples. Since $X(\omega)$ is periodic with period 2π. only samples in the fundamental frequency range are necessary. For convenience. we take N equidistant samples in the interval $0 \le \omega < 2\pi$ with spacing $\delta\omega = 2\pi/N$, as shown in Fig. 5.1. First, we consider the selection of N, the number of samples in the frequency domain.

if we evaluate (5.1.1) at $\omega = 2\pi k/N$, we obtain

$$X\left(\frac{2\pi}{N}k\right) = \sum_{n=-\infty}^{\infty} x(n)e^{-j2\pi kn/N} \qquad k = 0, 1, \ldots, N-1 \tag{5.1.2}$$

The summation in (5.1.2) can be subdivided into an infinite number of summations. where each sum contains N terms. Thus

$$\begin{aligned} X\left(\frac{2\pi}{N}k\right) = & \cdots + \sum_{n=-N}^{-1} x(n)e^{-j2\pi kn/N} + \sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N} \\ & + \sum_{n=N}^{2N-1} x(n)e^{-j2\pi kn/N} + \cdots \\ = & \sum_{l=-\infty}^{\infty} \sum_{n=lN}^{lN+N-1} x(n)e^{-j2\pi kn/N} \end{aligned}$$

If we change the index in the inner summation from n to $n - lN$ and interchange the order of the summation. we obtain the result

$$X\left(\frac{2\pi}{N}k\right) = \sum_{n=0}^{N-1} \left[\sum_{l=-\infty}^{\infty} x(n - lN)\right] e^{-j2\pi kn/N} \tag{5.1.3}$$

for $k = 0, 1, 2, \ldots, N-1$.

The signal

$$x_p(n) = \sum_{l=-\infty}^{\infty} x(n - lN) \tag{5.1.4}$$

obtained by the periodic repetition of $x(n)$ every N samples. is clearly periodic with fundamental period N. Consequently. it can be expanded in a Fourier

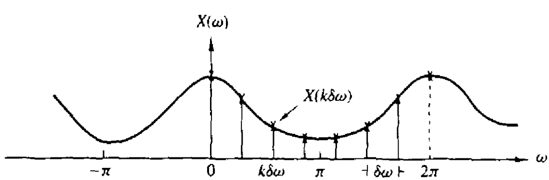


Figure 5.1 Frequency-domain sampling of the Fourier **transform**

series as

$$x_p(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N} \qquad n = 0, 1, \ldots, N-1 \tag{5.1.5}$$

with Fourier coefficients

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x_p(n) e^{-j2\pi kn/N} \qquad k = 0, 1, \ldots, N-1 \tag{5.1.6}$$

Upon comparing (5.1.3) with (5.1.6), we conclude that

$$c_k = \frac{1}{N} X\left(\frac{2\pi}{N}k\right) \qquad k = 0, 1, \ldots, N-1 \tag{5.1.7}$$

Therefore,

$$x_p(n) = \frac{1}{N} \sum_{k=0}^{N-1} X\left(\frac{2\pi}{N}k\right) e^{j2\pi kn/N} \qquad n = 0, 1, \ldots, N-1 \tag{5.1.8}$$

The relationship in (5.1.8) provides the reconstruction of the periodic signal $x_p(n)$ from the samples of the spectrum $X(\omega)$. However, it does not imply that we can recover $X(\omega)$ or $x(n)$ from the samples. To accomplish this, we need to consider the relationship between $x_p(n)$ and $x(n)$.

Since $x_p(n)$ is the periodic extension of $x(n)$ as given by (5.1.4), it is clear that $x(n)$ can be recovered from $x_p(n)$ if there is no aliasing in the time domain, that is, if $x(n)$ is time-limited to less than the period N of $x_p(n)$. This situation is illustrated in Fig. 5.2, where without loss of generality, we consider a finite-duration

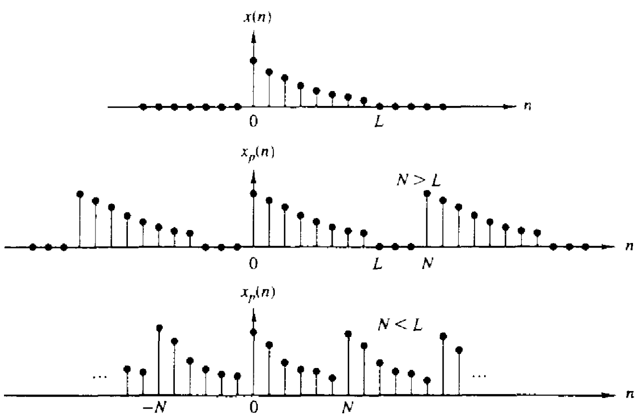


Figure 5.2 **Aperiodic sequence $x(n)$ of length L and its periodic extension for $N \geq L$ (no aliasing) and N $<$ L (aliasing).**

sequence $x(n)$, which is nonzero in the interval $0 \le n \le L-1$. We observe that when $N \ge L$,

$$x(n) = x_p(n) \qquad 0 \le n \le N-1$$

so that $x(n)$ can be recovered from $x_p(n)$ without ambiguity. On the other hand, if $N < L$, it is not possible to recover $x(n)$ from its periodic extension due to *time-domain aliasing*. Thus, we conclude that the spectrum of an aperiodic discrete-time signal with finite duration L, can be exactly recovered from its samples at frequencies $\omega_k = 2\pi k/N$, if $N \ge L$. The procedure is to compute $x_p(n)$, $n = 0, 1, \ldots, N-1$ from (5.1.8); then

$$x(n) = \begin{cases} x_p(n), & 0 \le n \le N-1 \\ 0, & \text{elsewhere} \end{cases} \tag{5.1.9}$$

and finally, $X(\omega)$ can be computed from (5.1.1).

As in the case of continuous-time signals, it is possible to express the spectrum $X(\omega)$ directly in terms of its samples $X(2\pi k/N)$, $k = 0, 1, \ldots, N-1$. To derive such an interpolation formula for $X(\omega)$, we assume that $N \ge L$ and begin with (5.1.8). Since $x(n) = x_p(n)$ for $0 \le n \le N-1$,

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X\left(\frac{2\pi}{N}k\right) e^{j2\pi kn/N} \qquad 0 \le n \le N-1 \tag{5.1.10}$$

If we use (5.1.1) and substitute for $x(n)$, we obtain

$$\begin{aligned} X(\omega) &= \sum_{n=0}^{N-1}\left[\frac{1}{N}\sum_{k=0}^{N-1} X\left(\frac{2\pi}{N}k\right) e^{j2\pi kn/N}\right] e^{-j\omega n} \\ &= \sum_{k=0}^{N-1} X\left(\frac{2\pi}{N}k\right)\left[\frac{1}{N}\sum_{n=0}^{N-1} e^{-j(\omega-2\pi k/N)n}\right] \end{aligned} \tag{5.1.11}$$

The inner summation term in the brackets of (5.1.11) represents the basic interpolation function shifted by $2\pi k/N$ in frequency. Indeed, if we define

$$\begin{aligned} P(\omega) &= \frac{1}{N}\sum_{n=0}^{N-1} e^{-j\omega n} = \frac{1}{N}\frac{1-e^{-j\omega N}}{1-e^{-j\omega}} \\ &= \frac{\sin(\omega N/2)}{N\sin(\omega/2)} e^{-j\omega(N-1)/2} \end{aligned} \tag{5.1.12}$$

then (5.1.11) can be expressed as

$$X(\omega) = \sum_{k=0}^{N-1} X\left(\frac{2\pi}{N}k\right) P\left(\omega - \frac{2\pi}{N}k\right) \qquad N \ge L \tag{5.1.13}$$

The interpolation function $P(\omega)$ is not the familiar $(\sin\theta)/\theta$ but instead, it is a periodic counterpart of it, and it is due to the periodic nature of $X(\omega)$. The phase shift in (5.1.12) reflects the fact that the signal $x(n)$ is a causal, finite-duration sequence of length N. The function $\sin(\omega N/2)/(N \sin(\omega/2))$ is plotted in Fig. 5.3 for $N = 5$. We observe that the function $P(\omega)$ has the property

$$P\left(\frac{2\pi}{N}k\right) = \begin{cases} 1, & k = 0 \\ 0, & k = 1, 2, \ldots, N-1 \end{cases} \tag{5.1.14}$$

Consequently, the interpolation formula in (5.1.13) gives exactly the sample values $X(2\pi k/N)$ for $\omega = 2\pi k/N$. At all other frequencies, the formula provides a properly weighted linear combination of the original spectral samples.

The following example illustrates the frequency-domain sampling of a discrete-time signal and the time-domain aliasing that results.

Example 5.1.1

Consider the signal

$$x(n) = a^n u(n) \qquad 0 < a < 1$$

The spectrum of this signal is sampled at frequencies $\omega_k = 2\pi k/N$, $k = 0, 1, \ldots, N-1$. Determine the reconstructed spectra for $a = 0.8$ when $N = 5$ and $N = 50$.

Solution The Fourier transform of the sequence $x(n)$ is

$$X(\omega) = \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \frac{1}{1 - ae^{-j\omega}}$$

Suppose that we sample $X(\omega)$ at N equidistant frequencies $\omega_k = 2\pi k/N$, $k = 0, 1, \ldots, N-1$. Thus we obtain the spectral samples

$$X(\omega_k) \equiv X\left(\frac{2\pi k}{N}\right) = \frac{1}{1 - ae^{-j2\pi k/N}} \qquad k = 0, 1, \ldots, N-1$$

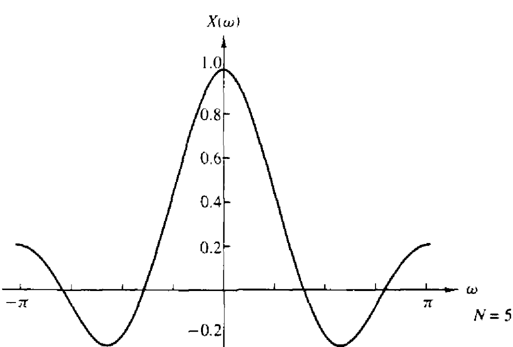


Figure 5.3 **Plot** of the function $[\sin(\omega N/2)]/[N \sin(\omega/2)]$.

The periodic sequence $x_p(n)$, corresponding to the frequency samples $X(2\pi k/N)$, $k = 0, 1, \ldots, N-1$, can be obtained from either (5.1.4) or (5.1.8). Hence

$$\begin{aligned} x_p(n) &= \sum_{l=-\infty}^{\infty} x(n-lN) = \sum_{l=-\infty}^{0} a^{n-lN} \\ &= a^n \sum_{l=0}^{\infty} a^{lN} = \frac{a^n}{1-a^N} \qquad 0 \le n \le N-1 \end{aligned}$$

where the factor $1/(1-a^N)$ represents the effect of aliasing. Since $0 < a < 1$, the aliasing error tends toward zero as $N \to \infty$.

For $a = 0.8$, the sequence $x(n)$ and its spectrum $X(\omega)$ are shown in Fig. 5.4a and b, respectively. The aliased sequences $x_p(n)$ for $N = 5$ and $N = 50$ and the corresponding spectral samples are shown in Fig. 5.4c and d, respectively. We note that the aliasing effects are negligible for $N = 50$.

If we define the aliased finite-duration sequence $x(n)$ as

$$\hat{x}(n) = \begin{cases} x_p(n), & 0 \le n \le N-1 \\ 0, & \text{otherwise} \end{cases}$$

then its Fourier transform is

$$\begin{aligned} \hat{X}(\omega) &= \sum_{n=0}^{N-1} \hat{x}(n)e^{-j\omega n} = \sum_{n=0}^{N-1} x_p(n)e^{-j\omega N} \\ &= \frac{1}{1-a^N} \cdot \frac{1-a^N e^{-j\omega n}}{1-ae^{-j\omega}} \end{aligned}$$

Note that although $\hat{X}(\omega) \neq X(\omega)$, the sample values at $\omega_k = 2\pi k/N$ are identical. *That* is,

$$\hat{X}\left(\frac{2\pi}{N}k\right) = \frac{1}{1-a^N} \cdot \frac{1-a^N}{1-ae^{-j2\pi kN}} = X\left(\frac{2\pi}{N}k\right)$$

5.1.2 The Discrete Fourier Transform (DFT)

The development in the preceding section is concerned with the frequency-domain sampling of an aperiodic finite-energy sequence $x(n)$. In general, the equally spaced frequency samples $X(2\pi k/N)$, $k = 0, 1, \ldots, N-1$, do not uniquely represent the original sequence $x(n)$ when $x(n)$ has infinite duration. Instead, the frequency samples $X(2\pi k/N)$, $k = 0, 1, \ldots, N-1$, correspond to a periodic sequence $x_p(n)$ of period N, where $x_p(n)$ is an aliased version of $x(n)$, as indicated by the relation in (5.1.4), that is,

$$x_p(n) = \sum_{l=-\infty}^{\infty} x(n-lN) \tag{5.1.15}$$

When the sequence $x(n)$ has a finite duration of length $L \le N$, then $x_p(n)$ is simply *a* periodic repetition of $x(n)$, where $x_p(n)$ over a single period **is**

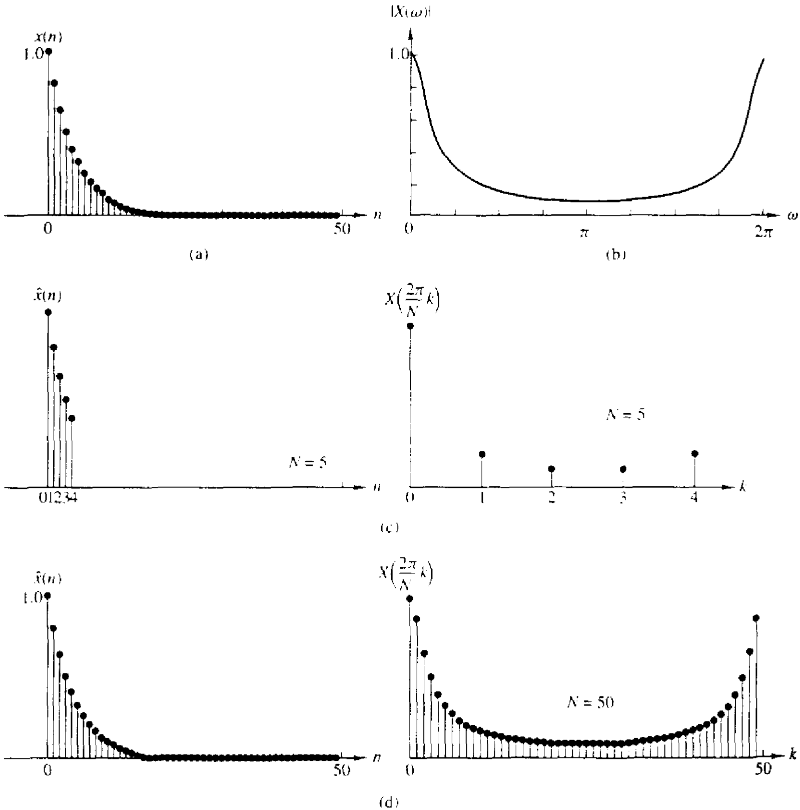


Figure 5.4 (a) Plot of sequence $x(n) = (0.8)^n u(n)$; *(b)* its Fourier transform **(magnitude** only); (c) effect of aliasing with $N = 5$; (d) reduced effect of aliasing with $N = 50$.

given as

$$x_p(n) = \begin{cases} x(n), & 0 \le n \le L - 1 \\ 0, & L \le n \le N - 1 \end{cases} \tag{5.1.16}$$

Consequently. the frequency samples $X(2\pi k/N)$, $k = 0, 1, \ldots, N - 1$, uniquely represent the finite-duration sequence $x(n)$. Since $x(n) \equiv x_p(n)$ over a single Period (padded by $N - L$ zeros). the original finite-duration sequence $x(n)$ can be obtained from the frequency samples $\{X(2\pi k/N)\}$ by means of the formula (5.1.8).

It is important to note that ***zero*** padding does not provide any additional information about the spectrum $X(\omega)$ of the sequence $\{x(n)\}$. The L equidistant samples of $X(\omega)$ are sufficient to reconstruct $X(\omega)$ using the reconstruction formula (5.1.13). However, padding the sequence $\{x(n)\}$ with N − L zeros and computing an N-point DFT results in a "better display" of the Fourier transform $X(\omega)$.

In summary, a finite-duration sequence $x(n)$ of length L [i.e., $x(n) = 0$ for $n < 0$ and $n \geq L$] has a Fourier transform

$$X(\omega) = \sum_{n=0}^{L-1} x(n)e^{-j\omega n} \qquad 0 \leq \omega \leq 2\pi \tag{5.1.17}$$

where the upper and lower indices in the summation reflect the fact that $x(n) = 0$ outside the range $0 \leq n \leq L - 1$. When we sample $X(\omega)$ at equally spaced frequencies $\omega_k = 2\pi k/N$, $k = 0, 1, 2, \ldots, N - 1$, where $N \geq L$, the resultant samples are

$$\begin{aligned} X(k) &\equiv X\left(\frac{2\pi k}{N}\right) = \sum_{n=0}^{L-1} x(n)e^{-j2\pi kn/N} \\ X(k) &= \sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N} \qquad k = 0, 1, 2, \ldots, N - 1 \end{aligned} \tag{5.1.18}$$

where for convenience. the upper index in the sum has been increased from $L - 1$ to $N - 1$ since $x(n) = 0$ for $n \geq L$.

The relation in (5.1.18) is a formula for transforming a sequence $\{x(n)\}$ of length $L \leq N$ into a sequence of frequency samples $\{X(k)\}$ of length **N**. Since the frequency samples are obtained by evaluating the Fourier transform $X(\omega)$ at a set of N (equally spaced) discrete frequencies. the relation in (5.1.18) is called the ***discrete Fourier transform*** (DFT) of $x(n)$. In turn. the relation **given** by (5.1.10). which allows us to recover the sequence $x(n)$ from the frequency samples

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{j2\pi kn/N} \qquad n = 0, 1, \ldots, N - 1 \tag{5.1.19}$$

is called the ***inverse*** DFT (IDFT). Clearly, when $x(n)$ has length $L < N$. the N-point IDFT yields $x(n) = 0$ for $L \leq n \leq N - 1$. To summarize, the formulas for the DFT and IDFT are

DFT

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N} \qquad k = 0, 1, 2, \ldots, N - 1 \tag{5.1.18}$$

IDFT

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{j2\pi kn/N} \qquad n = 0, 1, 2, \ldots, N - 1 \tag{5.1.19}$$

Example 5.1.2

A finite-duration sequence of length L is given as

$$x(n) = \begin{cases} 1, & 0 \le n \le L-1 \\ 0, & \text{otherwise} \end{cases}$$

Determine the N-point DFT of this sequence for $N \ge L$.

Solution The Fourier transform of this sequence is

$$X(\omega) = \sum_{n=0}^{L-1} x(n)e^{-j\omega n}$$

$$= \sum_{n=0}^{L-1} e^{-j\omega n} = \frac{1-e^{-j\omega L}}{1-e^{-j\omega}} = \frac{\sin(\omega L/2)}{\sin(\omega/2)} e^{-j\omega(L-1)/2}$$

The magnitude and phase of $X(\omega)$ are illustrated in Fig. 5.5 for $L = 10$. The N-point DFT of $x(n)$ is simply $X(\omega)$ evaluated at the set of N equally spaced frequencies $\omega_k = 2\pi k/N$, $k = 0, 1, \ldots, N-1$. I-fence

$$X(k) = \frac{1-e^{-j2\pi kL/N}}{1-e^{-j2\pi k/N}} \qquad k = 0, 1, \ldots, N-1$$

$$= \frac{\sin(\pi kL/N)}{\sin(\pi k/N)} e^{-j\pi k(L-1)/N}$$

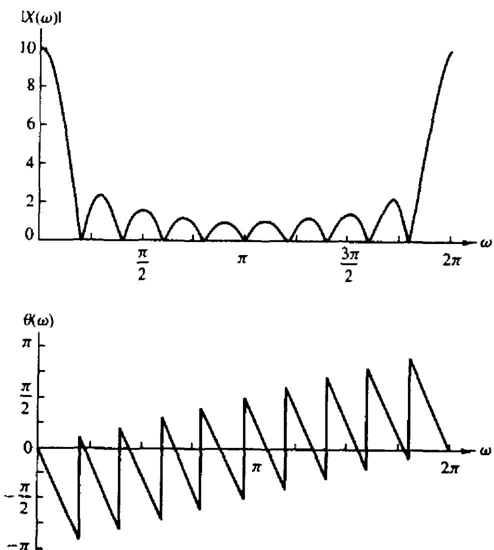


Figure 5.5 Magnitude and phase characteristics of the Fourier transform for signal in Example 5.12.

If N is selected such that $N = L$, then the DFT becomes

$$X(k) = \begin{cases} L, & k = 0 \\ 0, & k = 1, 2, \ldots, L-1 \end{cases}$$

Thus there is only one nonzero value in the DFT. This is apparent from observation of $X(\omega)$, since $X(\omega) = 0$ at the frequencies $\omega_k = 2\pi k/L$, $k \neq 0$. The reader should verify that $x(n)$ can be recovered from $X(k)$ by performing an L-point IDFT.

Although the L-point DFT is sufficient to uniquely represent the sequence $x(n)$ in the frequency domain, it is apparent that it does not provide sufficient detail to yield a good picture of the spectral characteristics of $x(n)$. If we wish to have better picture, we must evaluate (interpolate) $X(\omega)$ at more closely spaced frequencies, say $\omega_k = 2\pi k/N$, where $N > L$. In effect, we can view this computation as expanding the size of the sequence from L points to N points by appending $N - L$ zeros to the sequence $x(n)$, that is, zero padding. Then the N-point DFT provides finer interpolation than the L-point DFT.

Figure 5.6 provides a plot of the N-point DFT, in magnitude and phase, for $L = 10$, N = 50, and N = 100. Now the spectral characteristics of the sequence are more clearly evident, as one will conclude by comparing these spectra with the continuous spectrum $X(\omega)$.

5.1.3 The DFT as a Linear Transformation

The formulas for the DFT and IDFT given by (5.1.18) and (5.1.19) may be expressed as

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} \qquad k = 0, 1, \ldots, N-1 \tag{5.1.20}$$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn} \qquad n = 0, 1, \ldots, N-1 \tag{5.1.21}$$

where, by definition,

$$W_N = e^{-j2\pi/N} \tag{5.1.22}$$

which is an Nth root of unity.

We note that the computation of each point of the DFT can be accomplished by N complex multiplications and $(N - 1)$ complex additions. Hence the N-point DFT values can be computed in a total of N^2 complex multiplications and $N(N-1)$ complex additions.

It is instructive to view the DFT and IDFT as linear transformations on Sequences $\{x(n)\}$ and $\{X(k)\}$, respectively. Let us define an N-point vector $\mathbf{x}_N$ of the signal sequence $x(n)$, $n = 0, 1, \ldots, N-1$, an N-point vector $\mathbf{X}_N$ of frequency